

2024

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

SUPERVISOR HUNTING SYSTEM

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Supervisor Hunting System

Submitted to the Software Requirement Workshop

Bachelor of Computer Science (Software Engineering) with Honours

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1. INTRODUCTION

1.1 PURPOSE

The purpose of this document is to outline the specifications and requirements for the development of the Supervisor Hunting Application, designed to assist in the management of final year project (FYP) supervision for undergraduate students in the Faculty of Computing. The purpose of this document is to provide a comprehensive Software Requirements Specification (SRS) for the Final Year Project (FYP) Supervisor Hunting Application, an application proposed by the Faculty of Computing, University Malaysia Pahang Al-Sultan Abdullah. The aim of this document is to provide an overview of all parties associated with the project that all expected readers, which are developers, owner of the system (Faculty of Computing), respective stakeholders such as students, lecturers, Final Year Project coordinators, head of research group, and team members have to make sure that everyone is aware of their obligations and functions, it sets forth the objectives and requirements of the system.

The document encompasses the Software Requirements Specification (SRS) and the Request for Proposal (RFP). The SRS section ensures that the development team and stakeholders have a clear and common knowledge of what the system is designed to achieve by providing a full explanation of the functional and non-functional requirements of the system. In the RFP section, offers that can fulfil the requirements and goals of the project are welcomed, along with the criteria for choosing outside suppliers or partners, if necessary. This document's scope includes outlining project goals, like making it simpler for students can manage the supervisor hunting process and make student registration, the lecturer can post and manage interest FYP topics, manage appointment for student, the coordinators can manage the hunting schedule, announcement, manage quotas and generate supervisor hunting reports for coordinators. Then, other scopes are the system can manage notifications, giving real-time information on supervisor availability and Dean can view and monitor activities within the system.

1.2 SYSTEM IDENTIFICATION

The Software Requirement Specification (SRS) belongs to the “FYP Supervisor Hunting Application”.

Table 1.1 System Identification

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

System Identification Number	1000 (SRW_SRS_FYP_SHA 1.1.24) The system identification number shows the identification number for Final Year Project (FYP) Supervisor Hunting Application								
System Title	Final Year Project (FYP) Supervisor Hunting Application								
System Abbreviation	FYP_SHA								
System Version Number	1.1.24 This is initial release number for Final Year Project (FYP) Supervisor Hunting Application								
System Release Number	1.1.24 ➤ Start with number 1 as the major release number and middle with number 1 as the minor release number that this is initial version of the system. ➤ Number 24 as the service release that shows the system released in 2024.								
Document Identification ID	<i>SRS_FYP_SHA_2024</i> Meaning of term: <table><tr><td>SRS</td><td>SOFTWARE REQUIREMENT SPECIFICATION</td></tr><tr><td>FYP_SHA</td><td>Final Year Project Supervisor Hunting Application (name of system)</td></tr><tr><td>2024</td><td>Year of document created</td></tr></table>	SRS	SOFTWARE REQUIREMENT SPECIFICATION	FYP_SHA	Final Year Project Supervisor Hunting Application (name of system)	2024	Year of document created		
SRS	SOFTWARE REQUIREMENT SPECIFICATION								
FYP_SHA	Final Year Project Supervisor Hunting Application (name of system)								
2024	Year of document created								
Use Case ID	<i>MCAUC100_FYP_SHA_2024</i> Meaning of term: <table><tr><td>MCAUC</td><td>Manage Coordinator Activities Use Case</td></tr><tr><td>100</td><td>Number of use case in the system</td></tr><tr><td>FYP_SHA</td><td>Final Year Project Supervisor Hunting Application (name of system)</td></tr><tr><td>2024</td><td>Year of document created</td></tr></table> <i>MSUC200_FYP_SHA_2024</i> Meaning of term:	MCAUC	Manage Coordinator Activities Use Case	100	Number of use case in the system	FYP_SHA	Final Year Project Supervisor Hunting Application (name of system)	2024	Year of document created
MCAUC	Manage Coordinator Activities Use Case								
100	Number of use case in the system								
FYP_SHA	Final Year Project Supervisor Hunting Application (name of system)								
2024	Year of document created								

	MSUC	Manage Student Use Case	
	200	Number of use case in the system	
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)	
	2024	Year of document created	
	<i>MFYPUC300_FYP SHA_2024</i>		
	Meaning of term:		
	MSUC	Manage Final Year Project Use Case	
	300	Number of use case in the system	
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)	
	2024	Year of document created	
	<i>MSHUC400_FYP SHA_2024</i>		
	Meaning of term:		
	MSHUC	Manage Supervisor Hunting Use Case	
	400	Number of use case in the system	
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)	
	2024	Year of document created	

Sequence ID

MCABF101_FYPSHA_2024

Meaning of term:

MCASBF	Manage Coordinator Activities Basic Flow
101	Number of sequence diagram in the system
FYPSHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MCAAF102_FYPSHA_2024

Meaning of term:

MCAAF	Manage Coordinator Activities Alternative Flow
102	Number of sequence diagram in the system
FYPSHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MCAEF103_FYPSHA_2024

Meaning of term:

MCAEF	Manage Coordinator Activities Exception Flow
103	Number of sequence diagram in the system
FYPSHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MSBF201_FYPSHA_2024

Meaning of term:

MSBF	Manage Student Basic Flow
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	201	Number of sequence diagram in the system
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created
	<i>MSAF202_FYP SHA_2024</i>	
	Meaning of term:	
	MSAF	Manage Student Alternative Flow
	202	Number of sequence diagram in the system
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created
	<i>MSEF203_FYP SHA_2024</i>	
	Meaning of term:	
	MSEF	Manage Student Exception Flow
	203	Number of sequence diagram in the system
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created
	<i>MFYPBF301_FYP SHA_2024</i>	
	Meaning of term:	
	MFYPBF	Manage Final Year Project Basic Flow
	301	Number of sequence diagram in the system
	FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created
	<i>MFYP302_FYP SHA_2024</i>	

Meaning of term:

MFYPAF	Manage Final Year Project Alternative Flow
302	Number of sequence diagram in the system
FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MFYP303_FYP SHA_2024

Meaning of term:

MSHUC	Manage Final Year Project Exception Flow
303	Number of requirements in the system
FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MSHBF401_FYP SHA_2024

Meaning of term:

MSHBF	Manage Supervisor Hunting Basic Flow
401	Number of sequence diagram in the system
FYP SHA	Final Year Project Supervisor Hunting Application (name of system)
2024	Year of document created

MSHAF402_FYP SHA_2024

Meaning of term:

MSHAF	Manage Supervisor Hunting Alternative Flow
402	Number of sequence diagram in the system
FYP SHA	Final Year Project Supervisor Hunting

		Application (name of system)
	2024	Year of document created
	<i>MSHUC403_FYPSHA_2024</i>	
	Meaning of term:	
	MSHEF	Manage Supervisor Hunting Exception Flow
	403	Number of sequence diagram in the system
	FYPSHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created
Requirement Traceability ID	<i>ReqTrac100_FYPSHA_2024</i>	
	Meaning of term:	
	ReqTrac	Requirement Traceability
	100	Number of requirements in the system
	FYPSHA	Final Year Project Supervisor Hunting Application (name of system)
	2024	Year of document created

1.3 SYSTEM OVERVIEW

The proposed Supervisor Hunting App is designed to streamline the process of students in the Faculty of Computing finding supervisors for their Final Year Projects (FYP). This system aims to assist the FYP Coordinator in managing interactions between students and lecturers, ensuring an organized and efficient matching process based on project themes, student interests, and lecturer expertise.

The app serves various users: the Deputy Dean of Research, FYP Coordinator, Head of Research Group (HoRG), FC Lecturers, and FC Students. Key features include setting schedules and announcements for supervisor hunting activities, managing supervisor quotas, allowing lecturers to post FYP topics, enabling students to search and communicate with potential supervisors, and registering agreed supervisors. The system also provides real-time updates on supervisor availability, suggests relevant supervisors for students who haven't identified one, and allows the HoRG and Coordinator to monitor the process and generate reports.

Overall, the Supervisor Hunting App enhances efficiency, transparency, and communication in the supervisor selection process, ensuring every student finds a suitable supervisor for their FYP.

1.1 REFERENCES

This section shall list the number, title, revision, and date of all documents referenced in this specification. This section shall also identify the source for all documents not available through normal Government stocking activities.

2. PRODUCT DESCRIPTION

2.1 Product Perspective

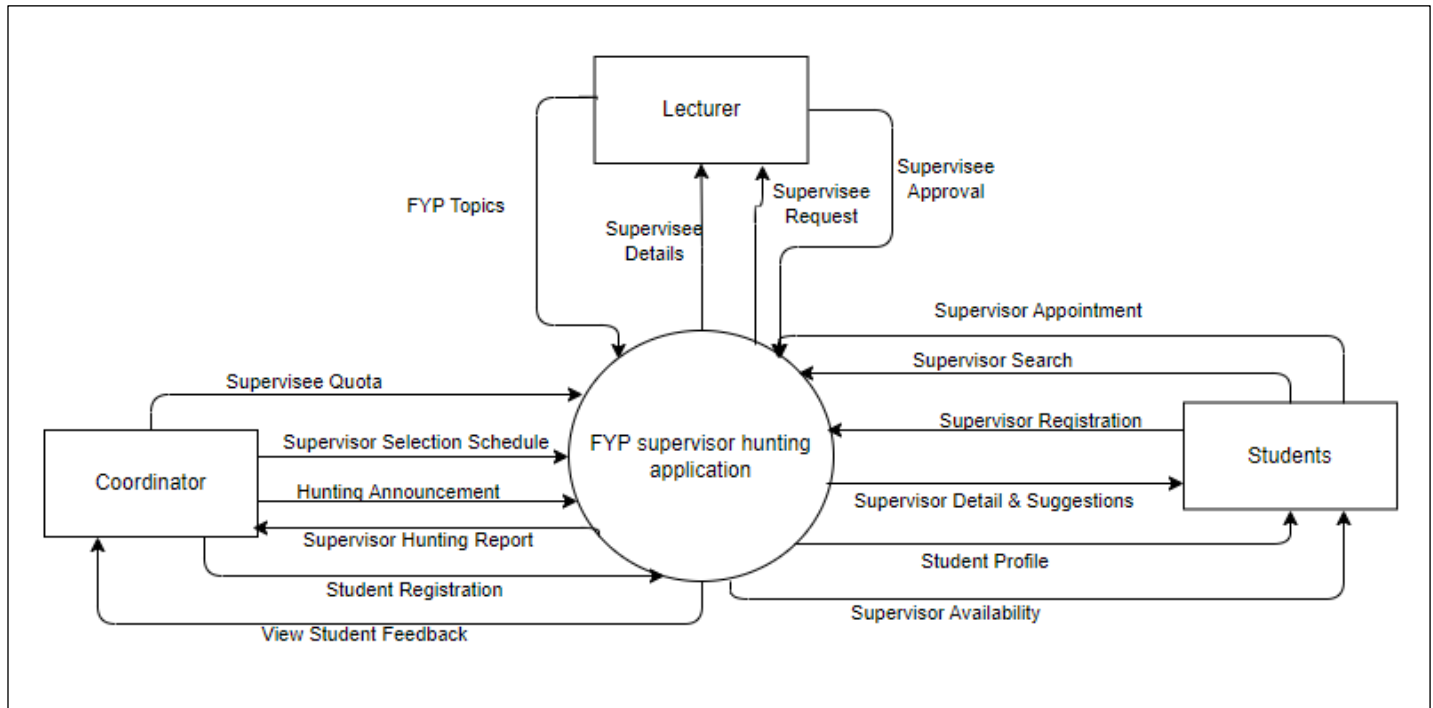


Figure 2.1 Context diagram of the Final Year Project (FYP) Supervisor Hunting Application

Illustrates the context diagram of the FYP Supervisor Hunting Application. A detailed description of the data flows is explained below.

1. Coordinator able to set the schedule and do the announcement regarding the supervisor hunting activities.

Data Flow:

- Supervisor Selection Schedule: The coordinator sends the schedule data to the system.
- Hunting Announcement: The coordinator sends announcements to the system.

Interaction: The coordinator inputs the schedule and announcements into the system, which disseminates this information.

2. Coordinator able to set and update the quota of supervisee for each lecturer.

Data Flow:

- Lecturer Quota: The coordinator updates lecturer quotas in the system.

Interaction: The coordinator manages the quota for each lecturer through the system.

3. Lecturers able to post their interest topic for Final Year Project (FYP).

Data Flow:

- Final Year Project (FYP) Topics Data: Lecturers input their FYP topics into the system.

Interaction: Lecturers use the system to enter their areas of interest for FYP.

4. Students are able to search and communicate with their prospective supervisor regarding their interest in Final Year Project (FYP)..

Data Flow:

- Supervisor Search: Students perform searches for prospective supervisors in the system.
- Student Message and Lecturer Message: Students and lecturers communicate through the system.

Interaction: Students use the system to search for supervisors and send messages to them. Lecturers respond via the system.

5. Students are able to register their agreed supervisor.

Data Flow:

- Agreed Supervisor Registration: Students register their chosen supervisors in the system.

Interaction: Once a student and supervisor agree, the student registers the selection in the system.

6. System should be able to update in real-time the available quota for each lecturer.

Data Flow:

- Implicit in the system's operations; real-time updates ensure accurate quota management based on student registrations.

Interaction: The system automatically updates lecturer quotas as students register their supervisors.

7. The system should be able to suggest any relevant supervisor or relevant research group for the student if the student is not able to indicate any prospective supervisor.

Data Flow:

- Supervisor Suggestions: The system suggests relevant supervisors to students.

Interaction: The system provides suggestions to students who do not have a

prospective supervisor based on their interests.

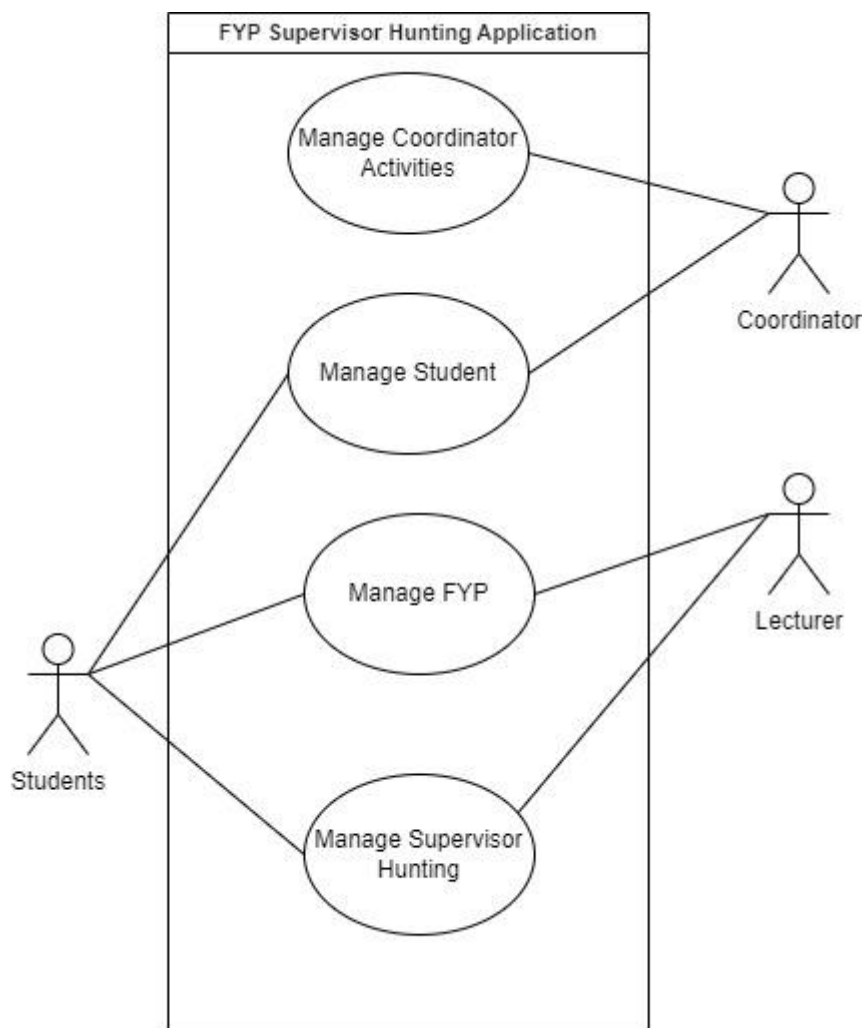
8. HORG and coordinator should be able to monitor supervisor hunting by generating relevant reports.

Data Flow:

- Supervisor Hunting Report: Generated by the system and accessible to the coordinator and HORG.
- Hunting Activity Monitoring: Sent to Dean for monitoring purposes.

Interaction: The system generates reports that HORG and coordinators can access to monitor the process.

2.2 Product Functions



2.3 User Characteristics

User	Educational Level	Experience	Technical Expertise
Final Year Project (FYP) Coordinator	Master's in Computer Science or a related field	Several years of experience in academic coordination, teaching, and student supervision.	Proficient in project management tools, student information systems, and coordination of academic programs.
Lecturer	Master's or PhD in Computer Science or a related field	Several years of teaching and supervising undergraduate and/or postgraduate projects.	Proficient in the subject matter they teach, experience with academic supervision, and familiarity with learning management systems.
Student	Undergraduate students in their final year of study.	Basic understanding of research methodologies and technical skills acquired during their FYP's.	Competence in relevant computing skills, familiarity with project management basics, and ability to use academic and communication platforms.

Table 2.1 User Characteristics of the stakeholders

3. SPECIFIC REQUIREMENTS

3.1 Software Product Features

3.1.1 Manage Coordinator Activities Use Case (Nur Asyikin)

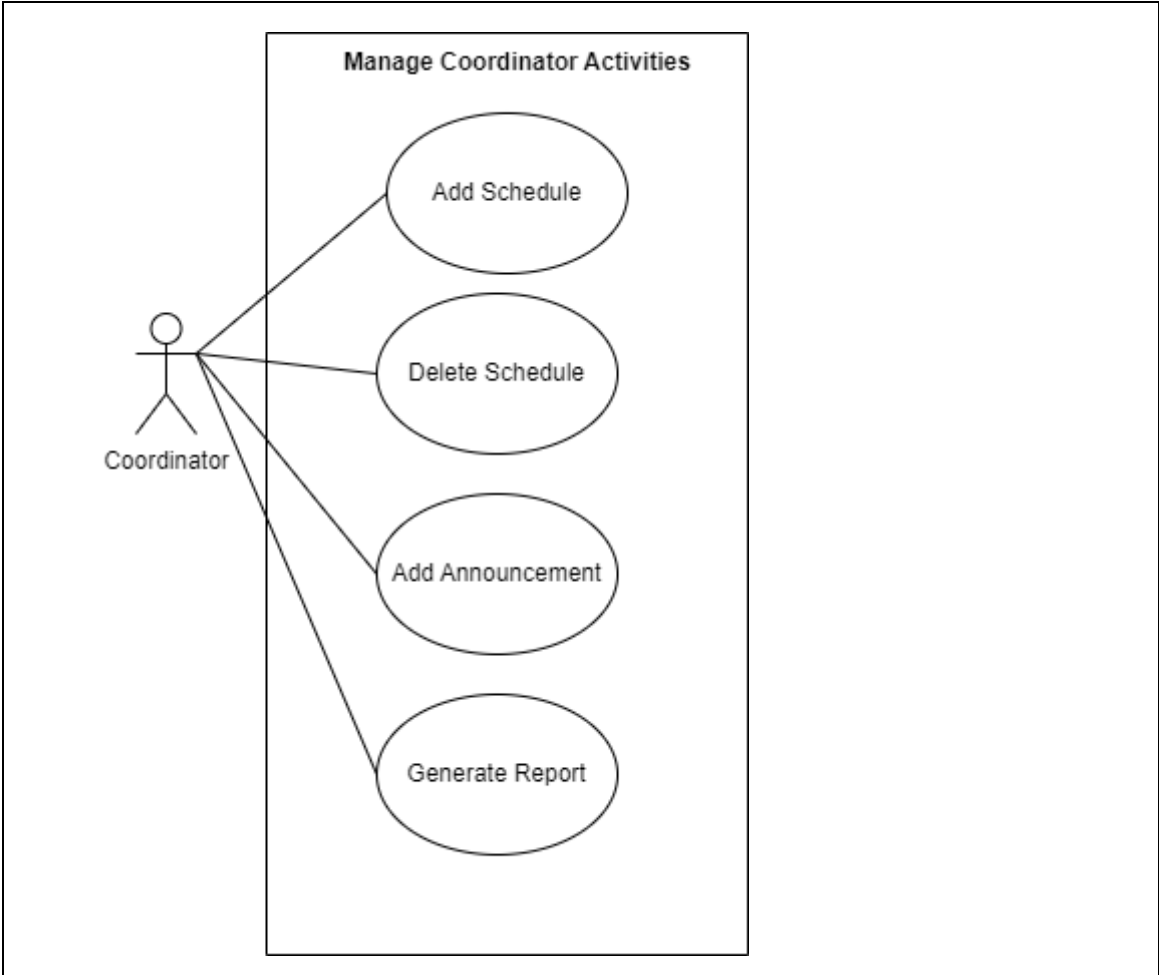


Figure 3.1 Manage Coordinator Activities Diagram

Use Case ID	MCAUC100_FYP_SHA_2024
Brief Description	This use case allows the coordinator to set schedules, making announcements. In addition, coordinator can generate report.
Actor	Coordinator
Pre-Conditions	1. The coordinator must be logged into the system.
Basic Flow	1. The Use case starts when the coordinator navigates to the Manage Coordinator activities dashboard.

	- The coordinator will see different buttons to choose which are [A1: Add Schedule] to set a schedule for supervisor hunting activities announcement. - The coordinator can go to schedule interface to delete the schedule [A2: Delete schedule]. - The coordinator can click button announcement to go to the announcement interface to add announcements [A3: Add announcement]. - The coordinator can click the button generate report to generate report [A4: generate the supervisor hunting report]. - The system Notify users that the schedule or announcement have been successfully updated via email. - The use case end.
Alternative Flow	Alternative Flow A1: Add Schedule - The system displays the schedule timeline interface. The coordinator is shown [A1. Add Schedule] by entering dates and inserting the time details to set the schedule. The coordinator can delete schedule if necessary [A2. Delete Schedule]. - The coordinator clicked button <<Add schedule>> to set the details, date and time chosen. TR1 - The system database saves the details, date and time. - The system prompt user that details are successfully update the message. - The use case continues at basic flow number 1. A2: Delete Schedule - The coordinator clicks the button <<Delete>> to delete the selected schedule. - The System processes the deletion request and deletes the schedule from the Database. - The Database saves the changes and confirms the deletion.

	- The Controller notifies the ManageSupervisorModule that the deletion was successful. - The ManageSupervisorModule informs the coordinator that the schedule has been successfully deleted. - The use case continues to step 1 in the Alternative Flow "Add Schedule". A3: Add Announcements - The coordinator clicks the button <<add announcements>> to add announcement. - The coordinator inputs the announcement details and duration of the announcement. - The coordinator clicks button <<Publish>> to display the announcement. TR2 - The database saves the changes of the system. - The system prompts a successfully published message. - The use case continues at basic flow number 1. A4: Generate the supervisor hunting report - The coordinator clicks the button <<Generate Report>> to Generate the supervisor hunting report. TR3 - The system displays the report. - The coordinator clicks button <<Save>> to save report file in device. - The database saves the changes of the system. - The system prompts a successfully save message. - The use case continues at basic flow number 1.
Exception Flow	Exception Flow E1: Report Generation Error - The system displays an error message indicating the failure. - The system logs the error details for troubleshooting. - The system prompts the user to try generating the report again through the message box. - The use case continues at step number 6 in alternative flow.
Post-Conditions	- The system notifies the coordinator that the schedule or announcements have been successfully updated via email. - The system shows a message box when the report had been save in the device.

Rules	1. Only users with Coordinator or HoRG roles can access this module.
Constraints	1. The announcement needs to be not more than 500 words per announcement.
Sequence Diagram	Refer Appendix A-2.1: Sequence Diagram – Basic Flow A-2.2: Sequence Diagram – Alternative Flow A-2.3: Sequence Diagram – Exception Flow

Table 3.1 Manage Supervisor Use Case Description

3.1.2 Manage Activities (Amala Karthigayan)

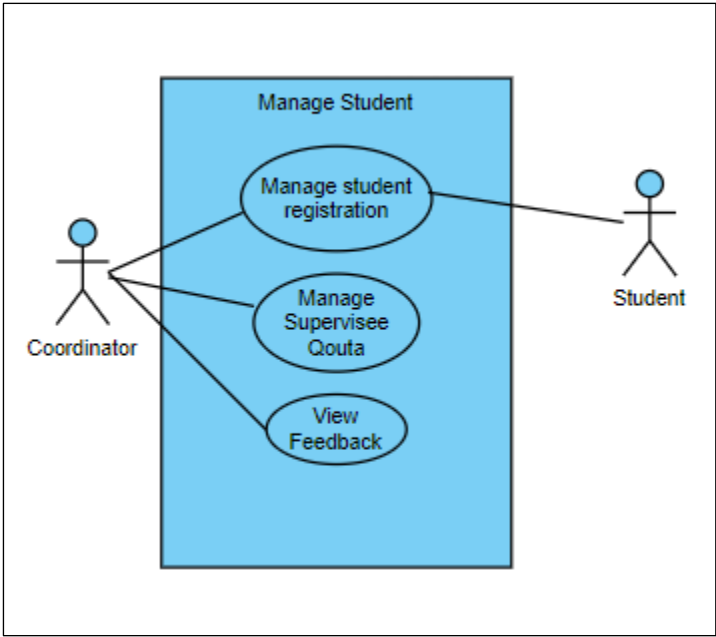


Table 3.2: Manage Activities - Use Case Description

Use Case ID	MSUC200_FYPSHA_2024
Brief Description	This use case allows the Coordinator to register students who have pre-registered for the Final Year Project (FYP) 1 subject, providing them with system access, manage supervisee quota for each lecturer and view feedback given by students about the system.
Actor	Coordinator and Student

Pre-Conditions	1. The user must be logged into the system as a Coordinator. 2. The Coordinator must have the document containing the list of students who have pre-registered for the FYP) 1 subject.
Basic Flow	Manage student registration (COORD_TRC_100): 1. Use case begins when the Coordinator logs into the system and the system displays the Coordinator's Dashboard. 2. The coordinator selects the <<Register Students>> button from the main menu. 3. The system displays the student registration page consists of the previous batches of registered students. 4. The coordinator has the option to: [A1: Create New Batch of Students] [A2: Update Batch] [A3: Delete Batch] 5. The coordinator clicks <<CONFIRM>> button. 6. The system displays all the batch icons based on the name given by the coordinator. 7. Coordinator clicks on the <<New Batch Icon>>. 8. System displays the new batch page and shows a "No students registered" message. 9. Coordinator clicks the [A4: Import Document] button to upload the list of pre-registered student details that has been accessed from the Open Registration system. 10. The system shows a list of students details imported from the document. 11. The Coordinator selects the number of students and clicks [A5: Generate Credentials] button to create unique UserID and passwords for the students. 12. The system notifies the students of their unique credentials via email. 13. The use case ends. Manage Supervisee Quota (COORD_TRC_200): 1. Use case begins when the Coordinator logs into the system and the system displays the Coordinator's Dashboard. 2. The coordinator selects the <<Manage Quota>> button from the main menu. 3. The system displays a list of supervisors and their respective supervisee quota. 4. The coordinator has an option to [A6: Add & Edit Quota]. 5. The coordinator clicks <<CONFIRM>> button. 6. Coordinator is able to view the current list of supervisors and their respective supervisee quota. 7. The use case ends.

	View Feedback (COORD_TRC_500): 1. Use case begins when the Coordinator logs into the system and the system display the Coordinator's Dashboard. 2. The coordinator selects the <<View Feedbacks>> button from the students. 3. System retrieves and displays the feedback list [E1: No Feedback Available]. 4. The coordinator reviews the feedback. 5. The system notifies the students after the feedback has been viewed by the coordinator. 6. The use case ends.
Alternative Flow	A1: Create New Batch of Students 1. System display a pop up page for the "New Batch Registration Form". 2. Coordinator fills in the details of the new batch [E2: Invalid Batch Name]. 3. Use case returns to step number 5 in Manage student registration. A2: Update Batch 1. Coordinator clicks on the <<Batch Icon>> that already has been created. 2. The system displays the selected Batch Page. 3. Coordinator clicks on the <<Update>> button. 4. Coordinator edit the batch details that she/he key-in before this. 5. Use case returns to step number 5 in Manage student registration. A3: Delete Batch 1. Coordinator clicks on the <<Batch Icon>> that already has been created. 2. The system displays the selected Batch Page. 3. Coordinator clicks on the <<Delete>> button. 4. The system displays a pop up confirmation message to delete the batch. 5. Use case returns to step number 5 in Manage student registration. A4: Import Document 1. System prompts the coordinator to upload the document containing list of pre-registered student details. 2. Coordinator upload the document [E3: Invalid File Format]. 3. System validates the data in the document [E4: Data not available].

	- Upon successful validation and document import, the system displays "File uploaded successfully" message. - Use case return to step number 10 in Manage student registration. A5: Generate Credentials - System generates a unique UserID and password for each pre-registered student in the list. - Upon successful credentials generation, the system displays "Credential generations successful" message. - Use case returns to step number 12 in Manage student registration. A6: Add & Edit Quota - Coordinator clicks on the <<Add & Edit Quota>> button for the selected supervisor. - Coordinator fills in the supervisee quota for the selected supervisor. - System update the quota details in the Quota database. - System displays a success message. - Use case returns to the step number 5 in Manage Supervisee Quota.
Exception Flow	E1: No Feedback Available - System displays a message indicating that "There's no feedback to display". - Coordinator goes back to dashboard. - Use case returns to step 6 in View Feedback. E2: Invalid Batch Name - The system validates the batch name entered by the Coordinator. - If the batch name is found to be invalid (e.g., it contains special characters, is too short or duplicates an existing batch name), the system displays an error message indicating the specific issue with the batch name. - The Coordinator corrects the batch name as per the guidelines provided by the system. - The system re-validates the corrected batch name. - Upon successful validation, the use case returns to step 3 in A1: Create New Batch of Students.

	E3: Invalid File Format 1. The system checks the format of the uploaded document. 2. If the file format is invalid (e.g., not a CSV or xlsx), the system displays an error message indicating the file format issue. 3. The Coordinator reviews the error message and corrects the file format as per the system requirements. 4. The Coordinator re-uploads the document. 5. The use case returns to step 3 in A4: Import Document. E4: Data not available 1. The system validates the data within the uploaded document. 2. If any required data is missing or incomplete (e.g., student names, IDs, etc.), the system displays an error message specifying the missing or incomplete data. 3. The Coordinator reviews the error message and updates the document to include the missing or incomplete data. 4. The Coordinator re-uploads the corrected document. 5. The use case returns to step 4 in A4: Import Document.
Post-Conditions	1. The student registration database is successfully updated and the students who are pre-registered for the FYP 1 subject are successfully registered in the system. 2. Supervisee quotas for every supervisor are successfully set and updated in the system and the database. 3. Notifications about feedback replies are sent to the respective students and lecturers via email.
Rules	1. Only students who are pre-registered for FYP 1 can be registered in the system. 2. Only users with Coordinator roles can access this use case.
Constraints	1. The system should validate the document format and data before processing. 2. Quota adjustments must not exceed the set limits (COORD_TRC_300) and

	should reflect the actual availability of supervisors in real time (COORD_TRC_400).
Sequence Diagram	Refer Appendix A-2.1: Sequence Diagram – Basic Flow A-2.2 :Sequence Diagram – Alternative Flow A-2.3 :Sequence Diagram – Exception Flow

3.1.3 Manage Final Year Project (Mirza Rusyaidi)

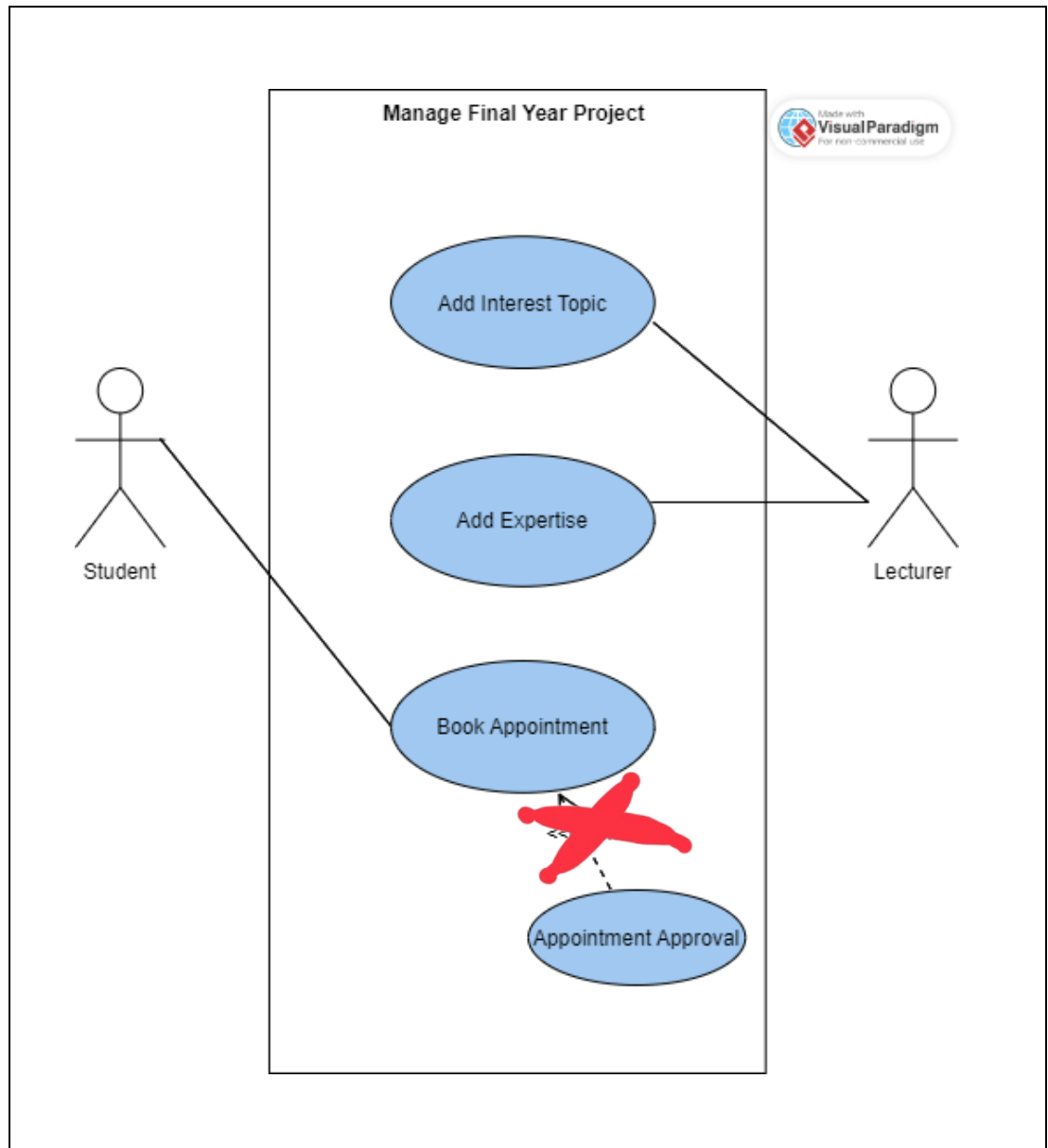


Figure 3.2 Manage Final Year Project Diagram

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Use Case ID	MFYPUC300_FYPSHA_2024
Brief Description	The Manage Final Year Project use case allows lecturer to add, update, view and delete their interest Final Year Project (FYP) topics and their expertise. Student can book appointment with the respective lecturer and lecturer can decide either to approve or reject the appointment.
Actor	Student, Lecturer
Pre-Conditions	1. The lecturer must be logged into the system as lecturer roles. 2. The student must be logged into the system as student roles.
Basic Flow	The use case begins with: Lecturer 1. The system display summary of the dashboard that contains registered supervisee, interest topics and supervisee request pending. 2. The lecturer need to update their expertise at profile page. ✗ 3. The system automatically updated the latest details of the lecturer at student dashboard. 4. The lecturer click on <<Registered Supervisee>> button. [A1: Show Registered Supervisee Details] ✗ 5. The lecturer click on <<Interest FYP Topics>> button. [A2: Show FYP Interest form] ✗ 6. The lecturer click on <<Supervisee Request Pending>> button. [A3: Show Supervisee Request Approval form] ✗ 7. The use case end. Student 1. The system shows Supervisor Details page. 2. The student clicks the <<Book Appointment>> button. 3. The system shows the Book Appointment form. 4. The student can fill in the Book Appointment form. 5. The student click on submit button. [E1: Cancel submit] 6. The system will automatically send a notification via email to the respective lecturer about the appointment request.

	7. The lecturer need to decide either to approve or reject the appointment. [A4: Appointment Approval]  5. The use case end.
Alternative Flow	[A1: Show Registered Supervisee Details] 1. The lecturer can view the details of registered supervisee. 2. The use case continue at number 1 in basic flow. [A2: Show FYP Interest form] 1. The lecturer can add FYP interest topics by fill in the FYP Interest Topic Form. 2. The lecturer clicks submit button. 3. The lecturer can make a change of the topic, if necessary, by clicks on FYP Interest Topic Update form. 4. The lecturer can delete the topic. 5. The use case continue at number 1 in basic flow. [A3: Show Supervisee Request Approval form] 1. The lecturer views pending supervisee request at Supervisee Request Pending page. 2. The lecturer can decide either to approve or reject the supervisee request. [A5: Supervisor Approval] 3. The use case continue at number 1 in basic flow. [A4: Appointment Approval] 1. The lecturer approved the request of appointment within 3 days working hours. [E2: Reject Request] 2. The system automatically send a notification via email to student regarding the appointment. 3. The use case continue at number 1 in basic flow. [A5: Supervisor Approval] 1. The lecturer approved the request if the student agree on the interest topic provide by the lecturer. [E3: Auto Cancelled] 2. The system automatically registered the student under selected

	supervisor. 3. The use case continue at number 1 in basic flow.
Exception Flow	[E1: Cancel submit] 1. The student click on cancel button. 2. The use case continue at number 1 in basic flow. [E2: Reject Request] 1. The lecturer reject the request. 2. The system send the notification via email regarding rejection status. 3. The use case end. [E3: Auto Cancelled] 1. The system auto cancelled the appointment if the lecturer did not respond within 3 working days. 2. The use case end.
Post-Conditions	1. The lecturer has successfully update interest FYP topics and update their expertise. 2. The student has book appointment with a desired lecturer to discuss regarding final year project. 3. Students should be able to view the expertise added by lecturers to choose a suitable supervisor.
Rules	1. Only lecturer can add FYP interest topic. 2. Both parties should receive a confirmation or rejection notification via email.
Constraints	1. Actions are restricted based on user roles (student or lecturer). 2. The student can book only one appointment at a time with a particular lecturer.
Sequence Diagram	Refer Appendix A-2.1: Sequence Diagram – Basic Flow A-2.2: Sequence Diagram – Alternative Flow A-2.3: Sequence Diagram – Exception Flow

3.1.4 Manage Supervisor Hunting (Nur Nabilah)

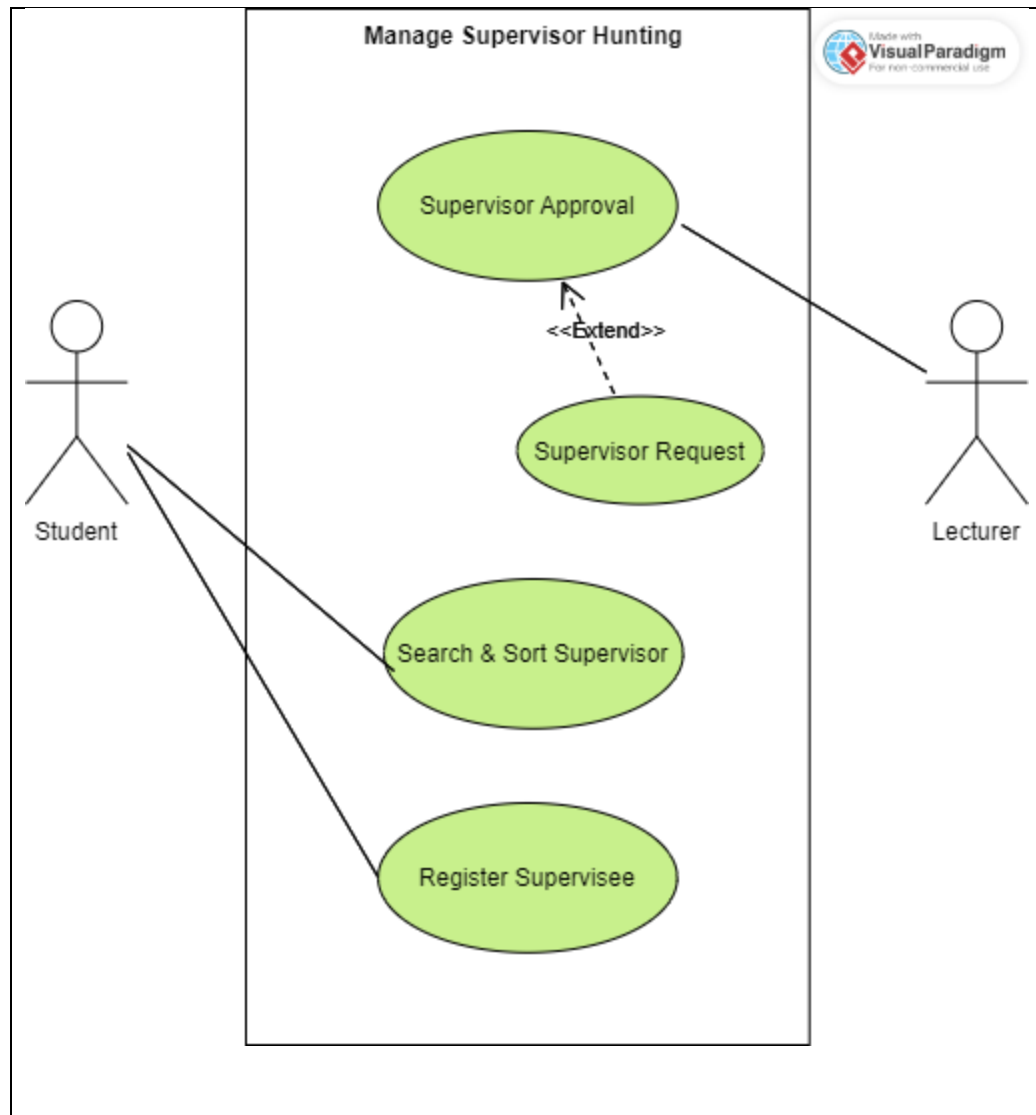


Figure 3.4 Manage Supervisor Hunting Diagram

Use Case ID	MSHUC400_FYP_SHA_2024
Brief Description	This use case allows the student to search, sort the research group interests and register a supervisor for their Final Year Project (FYP). Students can search for prospective supervisors and sort based on research group regarding their interest's topic for their Final Year Project (FYP) and submit requests to register as a supervisee. Lecturers can approve or reject these requests.
Actor	Student, Lecturer

Pre-Conditions	1. The student must be registered and logged into the system as “Student” role. 2. The lecturer must be registered and logged into the system as “Lecturer” role.
Basic Flow	The use case begins with: Student 1. The student redirect to the “Student Dashboard” section. 2. The student searches prospect supervisors by lecturer name. 3. The student enters name of lecture regarding their interest topic for Final Year Project (FYP). [E1: No Matching Supervisors Found] 4. The student sorts based on research group regarding their interest topic for Final Year Project (FYP). 5. The student selects any research group regarding their interest topic for Final Year Project (FYP). 6. The system displays a list of supervisors matching the student's interests. 7. The student selects a lecturer from the list. 8. The student fills out the information for request form such as propose FYP title and brief description. 9. The student clicks “Submit” button to register the chosen lecturer. 10. The system notified the student's request via email. 11. The use case ends. Lecturer 1. The lecturer redirect to the “Lecturer Dashboard” section. 2. The lecturer navigates to the “Supervisee Request Pending” page. 3. The system displays the list of pending requests registration from student. 4. The lecturer can decide either to approve or reject the supervisee request registration. [A1: Supervisee Registration] 5. The lecturer clicks “Save” button to save the application. 6. The system displays “Successfully Save” message. 7. The use case ends.
Alternative Flow	[A1: Supervisee Registration]

	1. The system notifies the student via email about supervisee application status of approved. [A2: Supervisee Rejection] 2. The system automatically registers the student as supervisee. 3. The process continues to step 7 from basic flow of Lecturer Flow. [A2: Supervisee Rejection] 1. The system notifies the student via email about supervisee application status of rejected and displays “Please Find Another Supervisor”. 2. The process continues to step 2 from basic flow of Student Flow.
Exception Flow	[E1: No Matching Supervisors Found] 1. The student may encounter no matching supervisors 2. The system displays “No Record Found” message. 3. The process continues to step 2 from basic flow of Student Flow.
Post-Conditions	1. The student has a registered supervisor for their Final Year Project (FYP). 2. The supervisor's list of students is updated to include the new student.
Rules	1. The student can only have one registered supervisor for their Final Year Project (FYP). 2. The lecturer can only approve requests up to their supervisory capacity such as number of students they can supervise. 3. Search can be filtered by interest topic, category and availability.
Constraints	1. Each lecturer can only supervise a limited number of students. 2. Requests should be submitted and approved or rejected within a specific time frame such as before the start of the semester. 3. The system must always be available and accessible to both students and lecturers, except during scheduled maintenance.
Sequence Diagram	A-4.1: Sequence Diagram – Basic Flow A-4.2: Sequence Diagram – Alternative Flow A-4.3: Sequence Diagram – Exception Flow

Table 3.4 Manage Supervisor Hunting Use Case Description

1.1 External Interface Requirements

1.1.1 User Interfaces

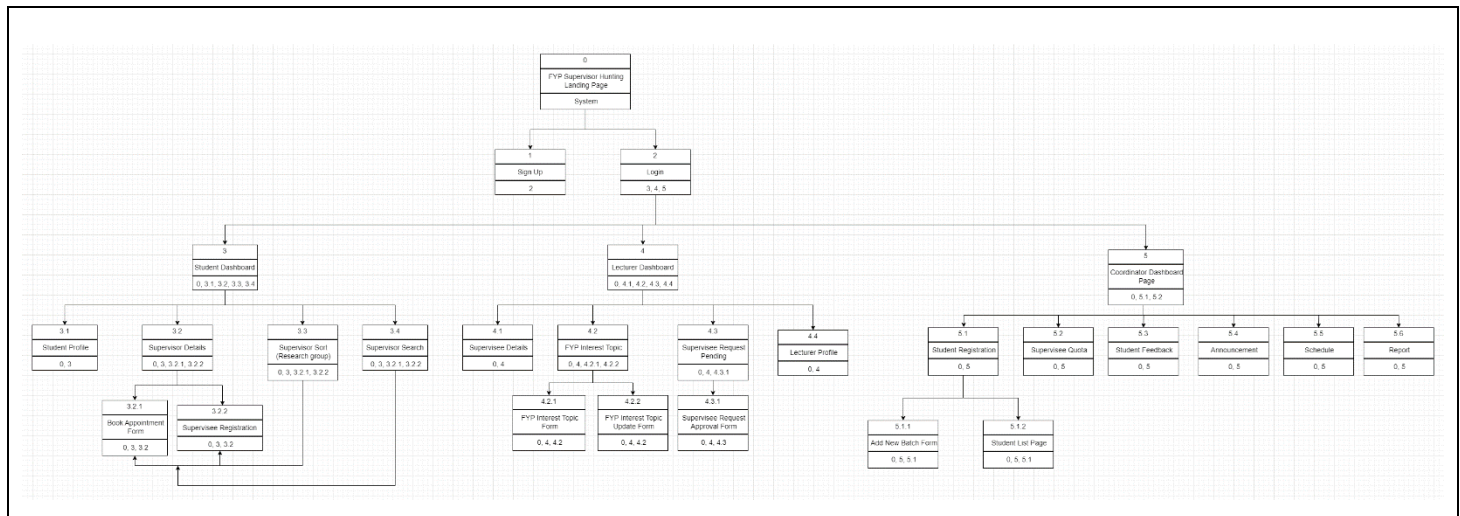


Figure 3.3 Dialogue Diagram

Table 3.3 User Interfaces Description

User Interface Name or Number	Description	User Interface Layout
Coordinator activities	From this page coordinator can go to add announcement, schedule, generate report	
Add announcement	The coordinator can add announcement details here	
Schedule	The coordinator can add schedule and delete schedule in this interface	
Generate Reports	The coordinator can generate report and save report file in this interface	

User Interface for Manage Student (Amala Karthigayan):

User Interface Name or Number	Description	User Interface Layout
Coordinator Dashboard	Displayed one the coordinator successfully logs into the system. It provides access to various navigations such as Manage Student Registration, Manage Supervisee Quota and View Feedback.	Refer Appendix B.2
Student Registration Page	- Displays the existing batches of students registered for the FYP 1 subject. <<Register New Batch>> button: Display the add new batch form.	
Add New Batch Form	Allows the coordinator to create a new batch by filling in the batch details.	
Student List Page	<<Update Batch>> button: Allows the coordinator to update batch name. <<Delete Batch>> button: Allows the coordinator to delete the selected batch. <<Import Document>> button: Allows coordinator to upload the CSV or xlsx document containing list of pre-registered student details. <<Generate Credential>> button: System creates unique UserID and password for every pre-registered student.	

	Display registered student list with their generated unique UserID and password.	
Supervisee Quota	- Displays a list of active supervisors. - Allows the Coordinator to manage (set & update) the supervisee quotas for each lecturer.	
View Feedback	Allows the Coordinator to view feedbacks and mark the feedbacks that has been viewed from students about the system.	

User Interface for Manage Final Year Project (MIRZA RUSYAUDI BIN RAHAMAT):

User Interface Name or Number	Description	User Interface Layout
Lecturer Dashboard	Displayed the Lecturer Dashboard that have three main function which is Registered Supervisee, FYP Interest Topic and Supervisee Request Pending.	Refer Appendix B.3
Supervisee Details	Displayed all the details of supervisee which is name, matric ID and FYP title.	
FYP Interest Topic	Displayed all the FYP interest topic. At this page, lecturer can edit, view and delete the topic.	
Supervisee Request Pending	Displayed all the pending request by supervisee.	
Supervisor Profile	Displayed all the lecturer info which is name, staff id, email, phone number and expertise.	
Book Appointment	Displayed the lecturer details and have book appointment form if the student want to make a appointment with the selected lecturer.	

User Interface for Manage Supervisor Hunting (NABILAH BINTI MOHAMMAD SHAHIDAN):

User Interface Name or Number	Description	User Interface Layout
Student Dashboard	Displayed the Student Dashboard that successfully logs into the system. It allows the student to search, sort the research group interests and register a supervisor for their Final Year Project (FYP). Students can search for prospective supervisors and sort based on research group regarding their interest's topic for their Final Year Project (FYP) and submit requests to register as a supervisee.	Refer Appendix B.4
Student Search	- Searches by lecturer name as their prospect supervisor. - Display list of supervisors.	
Student Sort	Allows the student to sort their interest topic Final Year Project based on research group.	
Supervisee Registration Page	<<Submit>> button: Allows the student to register their chosen supervisor.	
Lecturer Dashboard	Display the Lecturer Dashboard that successfully logs into the system. It allows the lecturers to approve or reject these requests.	
Supervisee Request Pending Page	Display the list of supervisee request pending.	
Supervisee Request Approval Form Page	<<Status>> dropdown button: Allows the lecturer either to approve or reject the request supervisor application.	

1.1.2 Hardware Interface

Not applicable.

1.1.3 Software Interface

Table 3.5 Software Interface

Application	API interface and Descriptions
Google Calendar API	Use through direct HTTP calls or with Google Client Libraries. It provides access to most of the features found in the Google Calendar Web interface. Will be used in Coordinator Dashboard.

2. REQUIREMENT TRACEABILITY

Table 4.1 Requirement Traceability for Use Case Manage Coordinator Activities (Nur Asyikin)

<i>Requirement ID</i>	<i>Description</i>	<i>Interface ID</i>
ReqTrac101_FYPSH A_2024	<<Add schedule>> button The coordinator able to add schedule for announcement.	Refer Appendix B.1
ReqTrac102_FYPSH A_2024	<<Publish>> The coordinator able to add announcement to display on the website for all to see.	
ReqTrac103_FYPSH A_2024	<<Generate Report >> button. The coordinator can monitor the system activities by generating reports. The coordinator can click the <<save>> button to save the generated report.	

Table 4.2 Requirement Traceability for Use Case Manage Student (Amala Karthigayan)

<i>Requirement ID</i>	<i>Requirement Description</i>	<i>Traceability ID</i>	<i>Traceability Description</i>	<i>Interface ID</i>
COORD_REQ_100	Coordinator registers students who have pre-registered for the FYP 1 course, providing them with system access	COORD_TRC_100	<<Register Student>> button <<Add New Batch>> button <<Import Document>> button <<Generate Credentials>> button	Refer Appendix B.2

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

COORD_REQ_200	Coordinator able to set and update the quota of supervisee for each lecturer.	COORD_TRC_200	<<Manage Quota>> button	
COORD_REQ_300	Each lecturer is not allowed to receive more than their assigned quota for the students.	COORD_TRC_300	Quota Limit Validation	
COORD_REQ_400	System should be able to update in real time the available quota for each lecturer	COORD_TRC_400	Real-Time Quota Update Function	
COORD_REQ_500	Any suggestion(s) function to increase the efficiency of this application are most welcome.	COORD_TRC_500	<<View Feedback>> button The Coordinator clicks the <<View Feedback>> button to view feedback submitted by students.	
COORD_REQ_600	Easy to use and require minimum experience to learn the system for first-time users.	COORD_TRC_600	User Friendly Interface Design	

Table 4.3 Requirement Traceability for Use Case Manage Final Year Project (MIRZA
RUSYAI DI BIN RAHAMAT)

<i>Requirement ID</i>	<i>Requirement Description</i>	<i>Traceability ID</i>	<i>Traceability Description</i>	<i>Interface ID</i>
LECT_REQ_300	Lecturer allows to add FYP interest topic.	LECT_TRC_300	Fill in form <<FYP Interest Topic>>	Refer Appendix B.3
LECT_REQ_301	Lecturer allows to update their expertise at lecturer profile.	LECT_TRC_301	Update Lecturer Profile form <<Expertise>>	
STUD_REQ_302	Student allows to make appointment with desired lecturer.	STUD_TRC_302	Fill in appointment form <<Date, Time, Reason>>	

Table 4.3 Requirement Traceability for Use Case Manage Supervisor Hunting (NABILAH
BINTI MOHAMMAD SHAHIDAN)

<i>Requirement ID</i>	<i>Requirement Description</i>	<i>Traceability ID</i>	<i>Traceability Description</i>	<i>Interface ID</i>
STUD_REQ_400	Student allows to search for prospective supervisors by searching lecturer name.	STUD_TRC_400	Fill out search box <<Lecturer Name>>	Refer Appendix B.3
STUD_REQ_401	Student sorts the research group interests for their Final Year Project (FYP).	STUD_TRC_401	Sorts drop down button <<Research Group>> Display <<List of Lecturer>>	
STUD_REQ_402	Student registers a lecturer as supervisor by submitting the register form.	STUD_TRC_402	Fill out register form <<Matric ID, Title, Brief Description>> Click <<Submit>>	

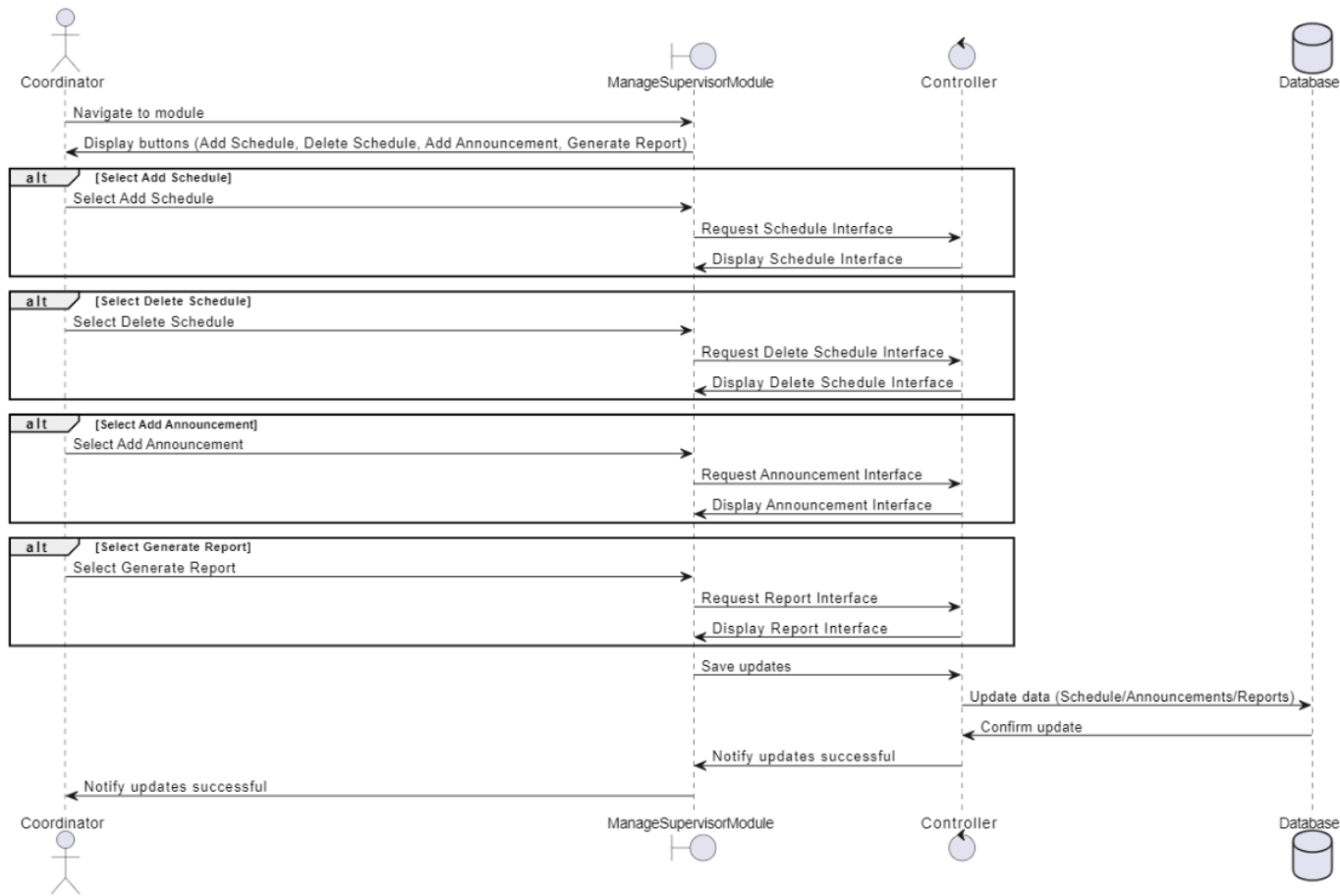
			button.	
LECT_REQ_403	Lecturers can approve or reject these requests.	LECT_TRC_403	Approval Validation	
LECT_REQ_400	System should be able to update in real time the available quota for each lecturer	LECT_TRC_400	Real-Time Quota Update Function	
COORD_REQ_600	Easy to use and require minimum experience to learn the system for first-time users.	COORD_TRC_600	User Friendly Interface Design	

3. ACRONYMS AND ABBREVIATION

APPENDIX A
Sequence Diagram

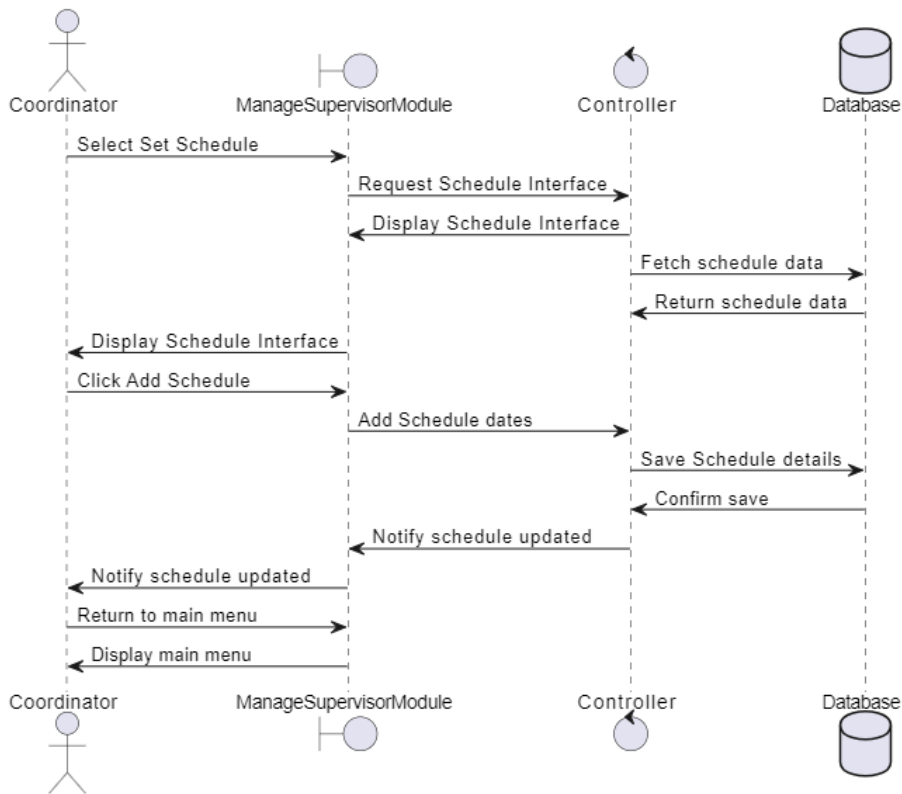
1. Manage Coordinator Activities

1.1 Basic Flow (MCABF101_FYPSHA_2024)

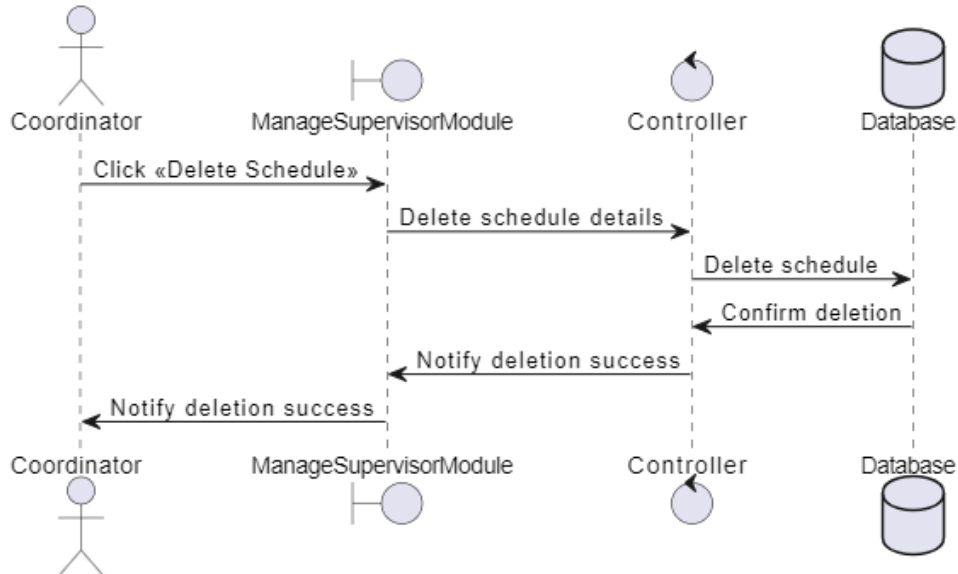


1.2 Alternative Flow (MCAAF102_FYPSHA_2024)

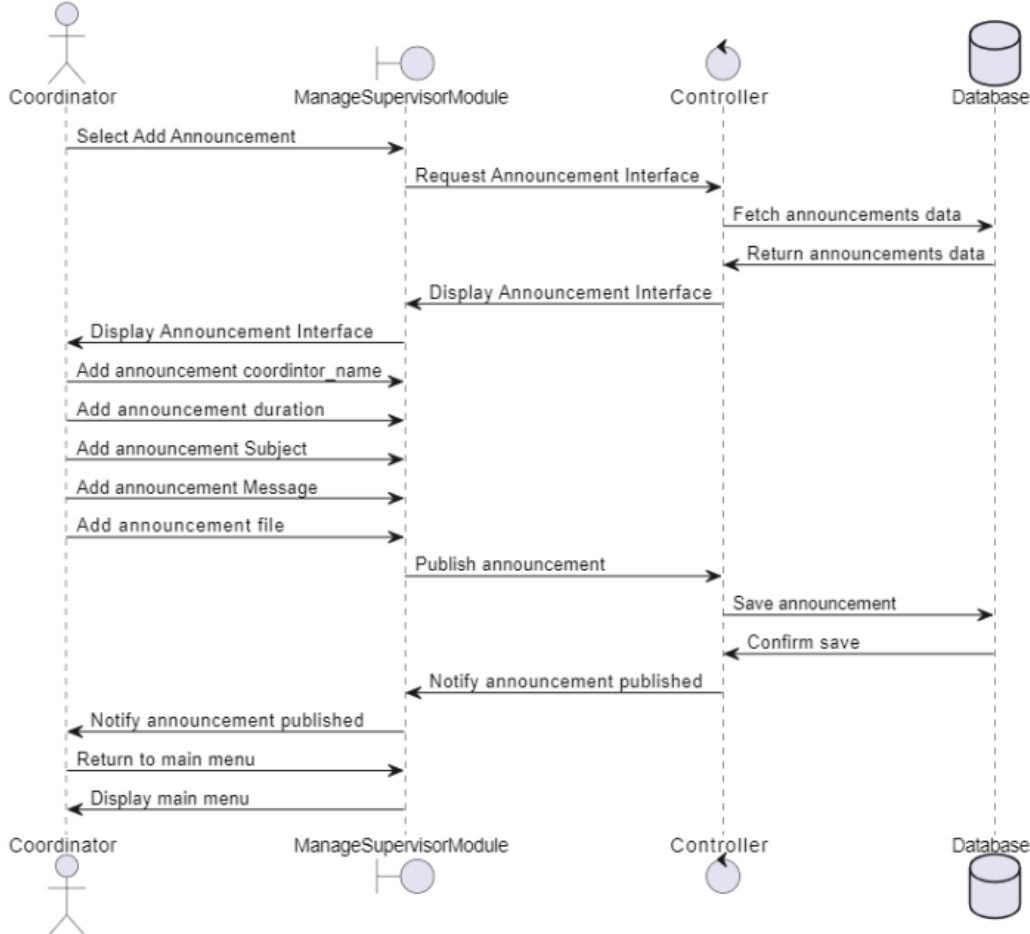
- Add Schedule



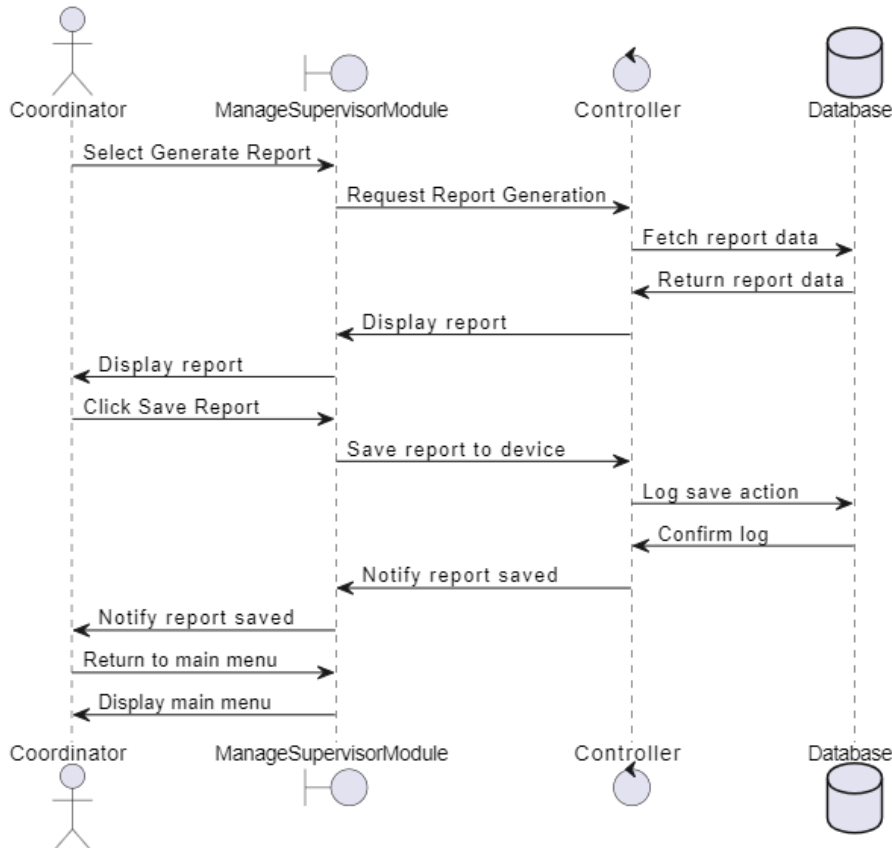
- Delete Schedule



- Add announcement

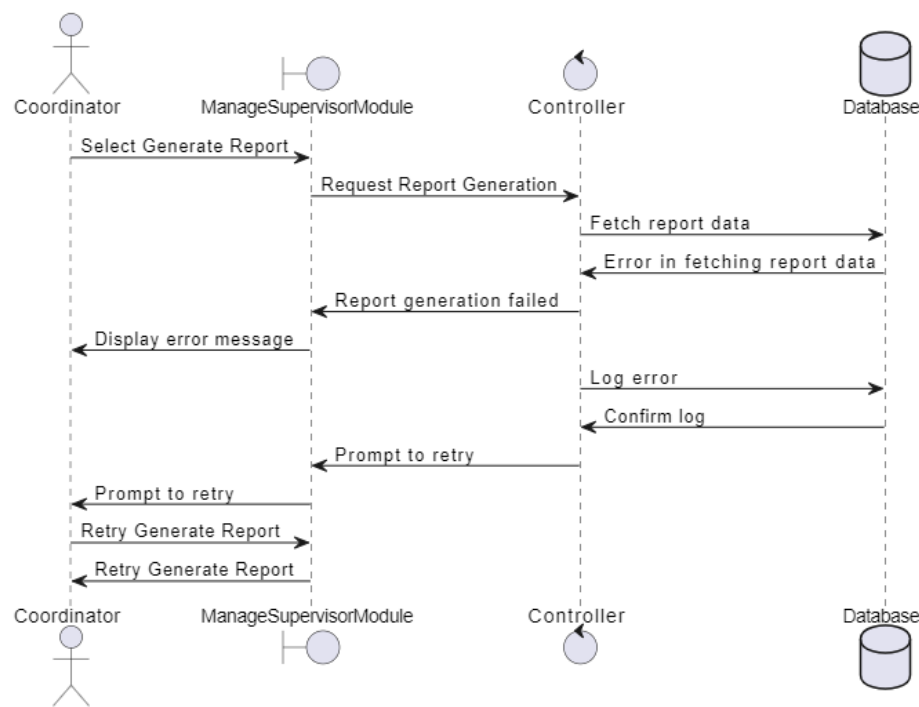


• Generate Report



1.3 Exception Flow (MCAEF103_FYPSHA_2024)

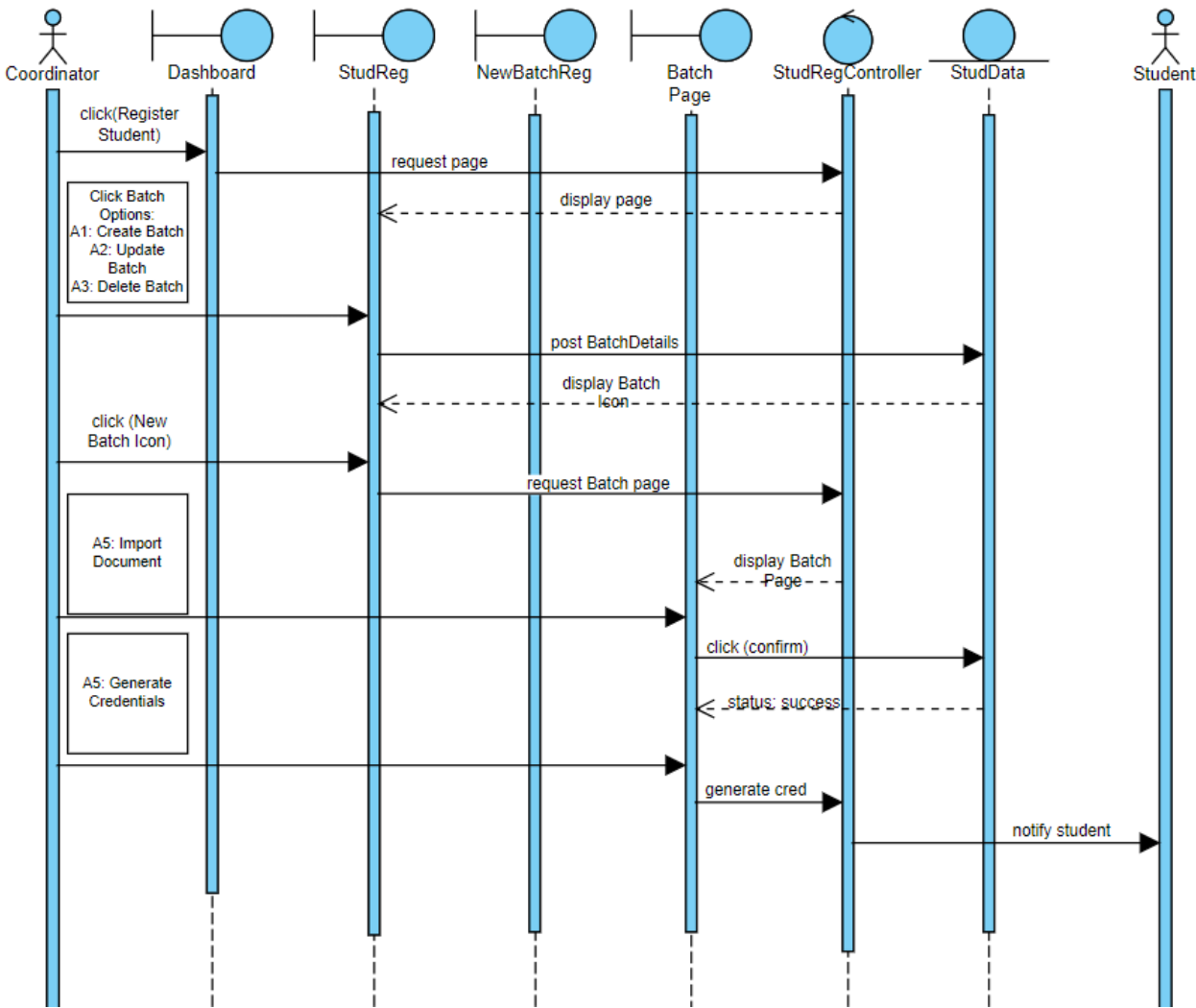
- Report Generation error



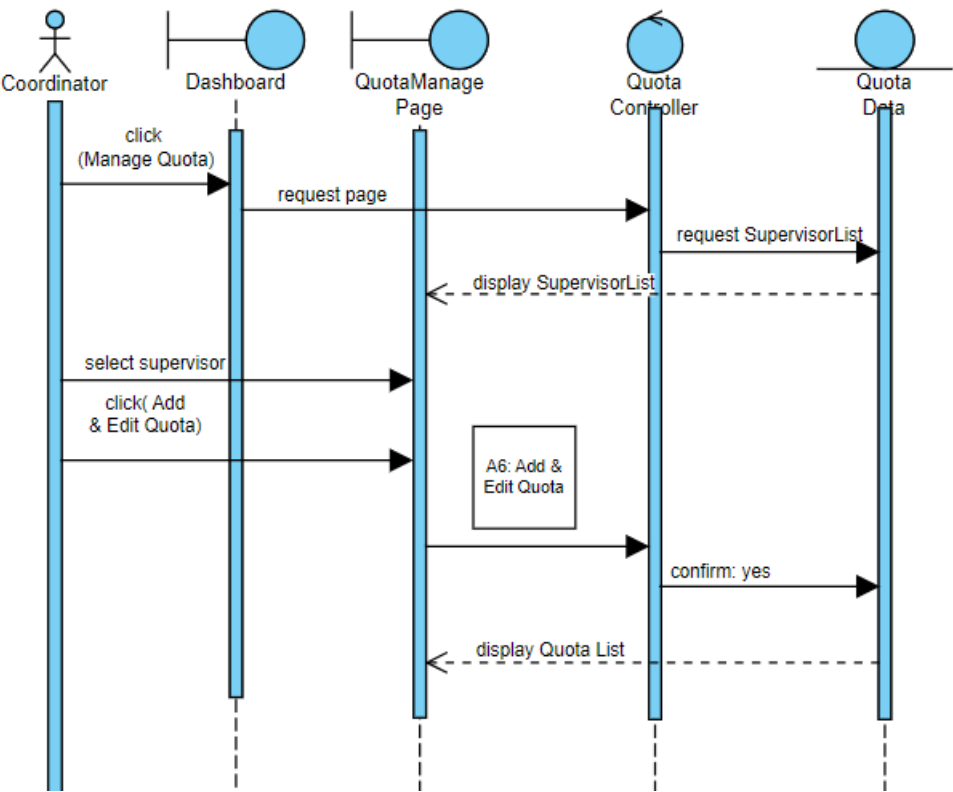
2. Manage Student (Amala Karthigayan)

2.1 Basic Flow (MSBF201_FYPSHA_2024)

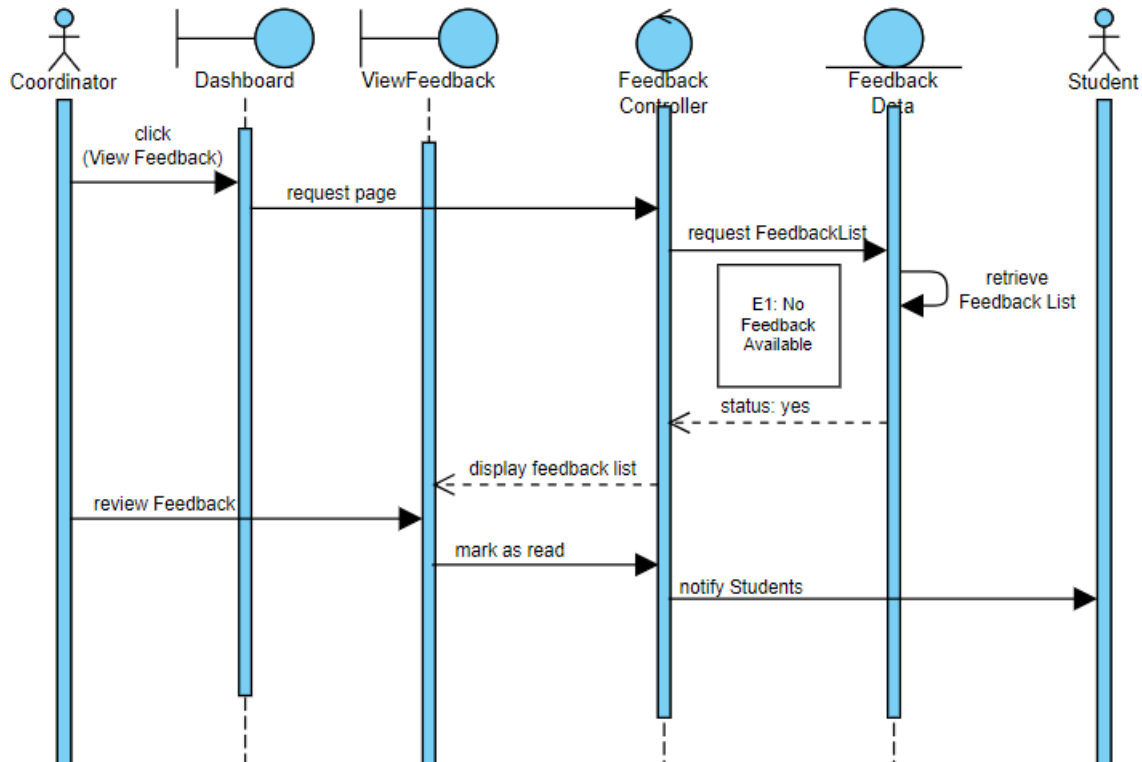
- Manage Student Registration



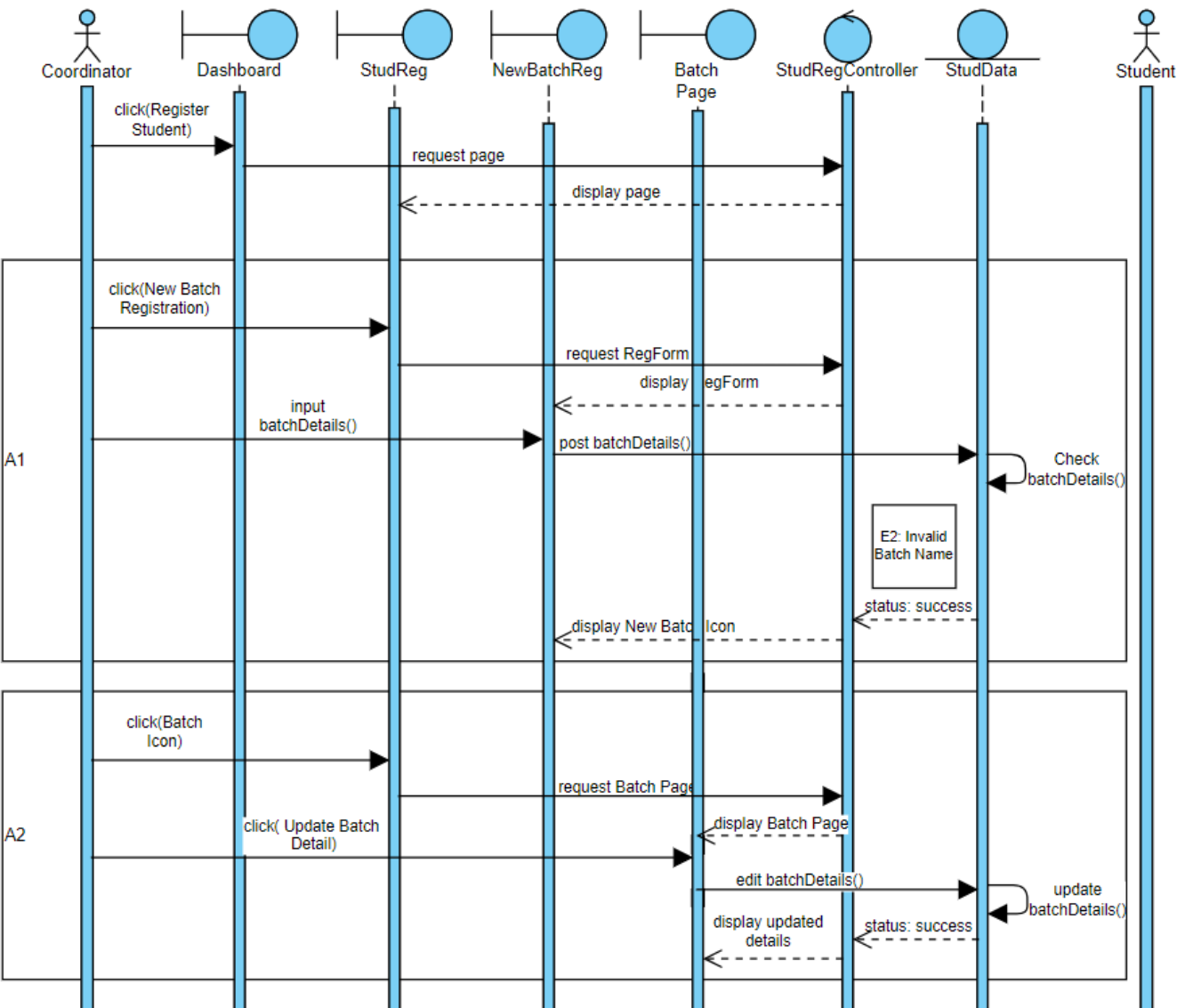
Manage Supervisee Quota

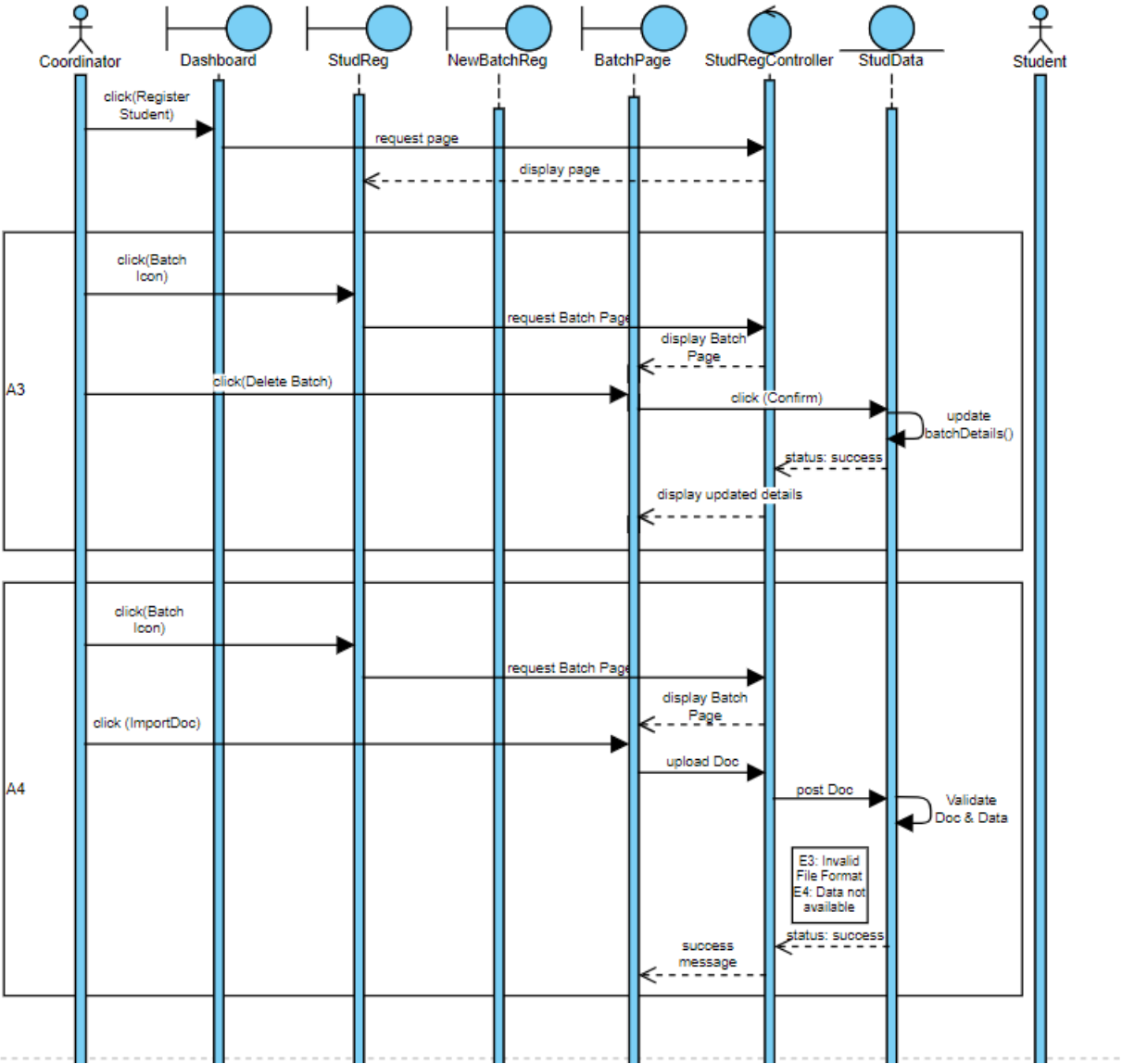


- View Feedback

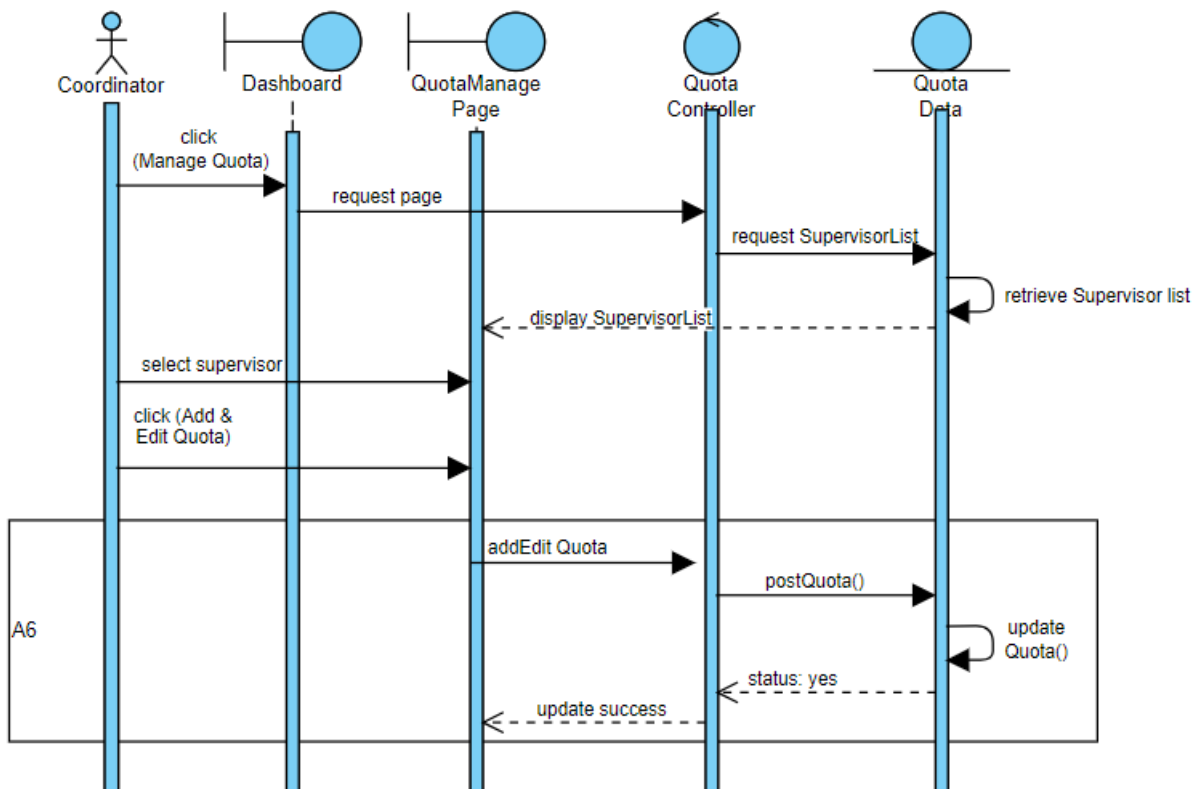
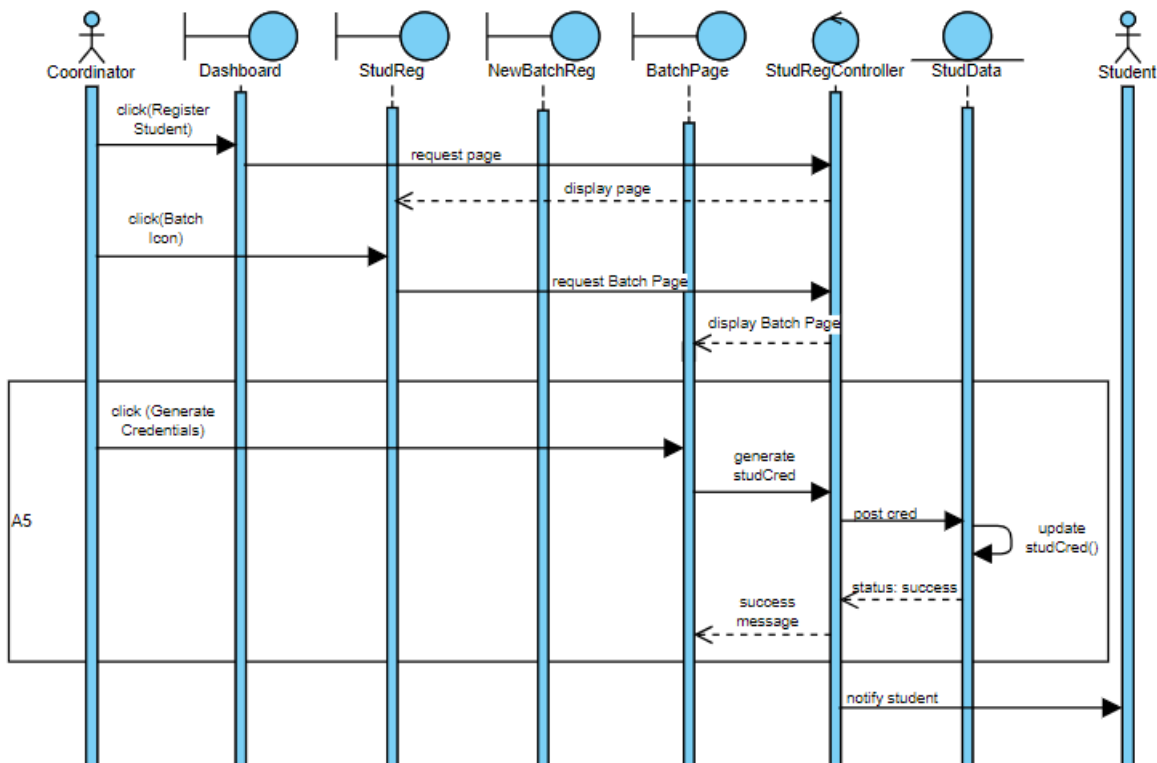


2.2 Alternative Flow (MSAF202_FYPSHA_2024)

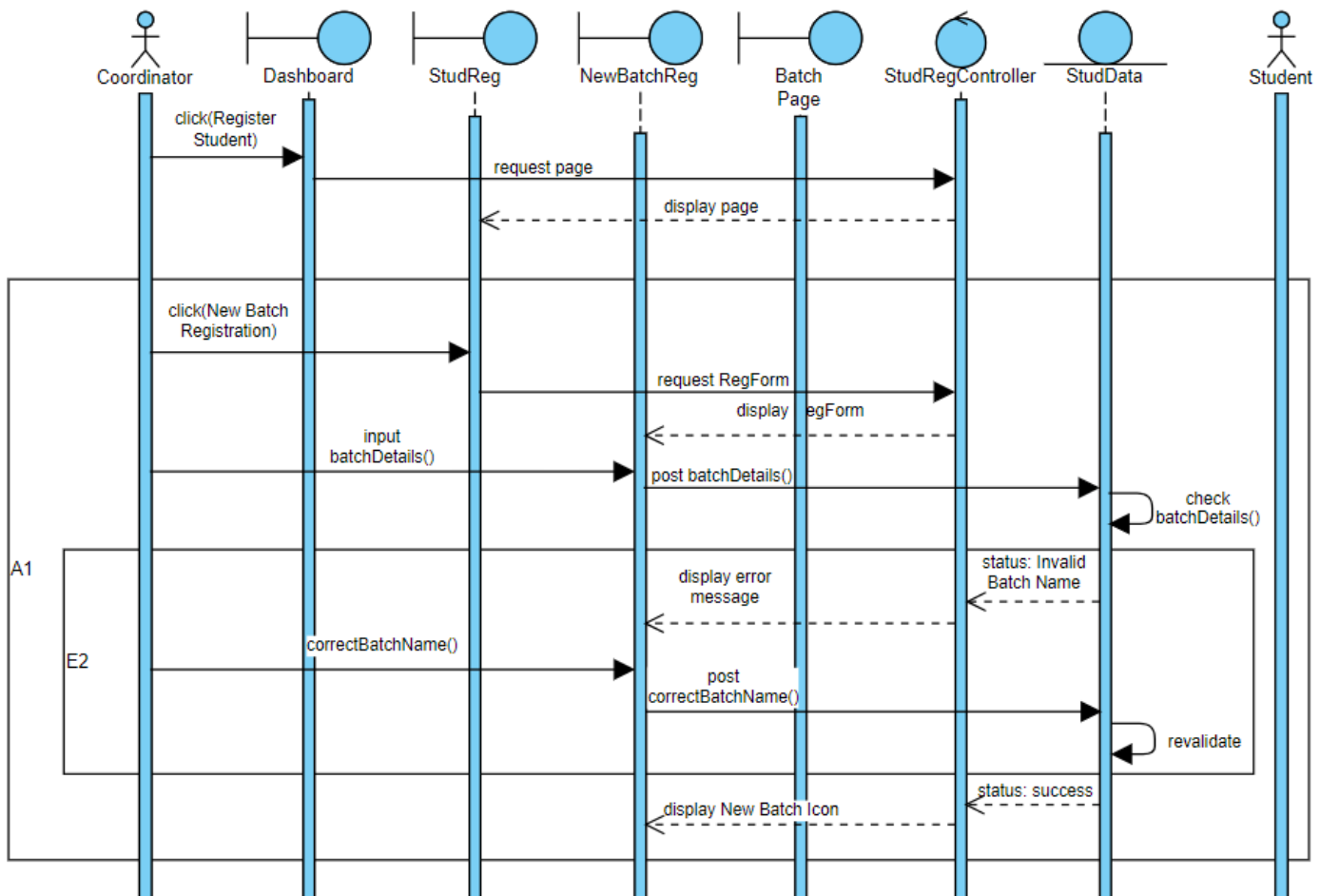
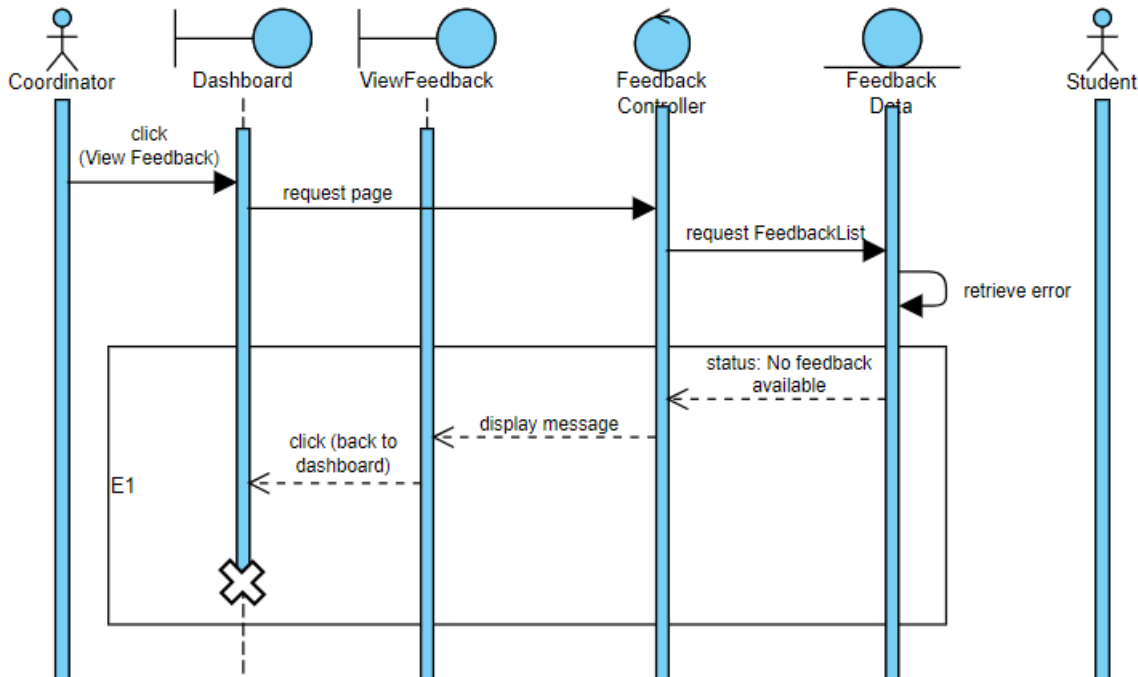




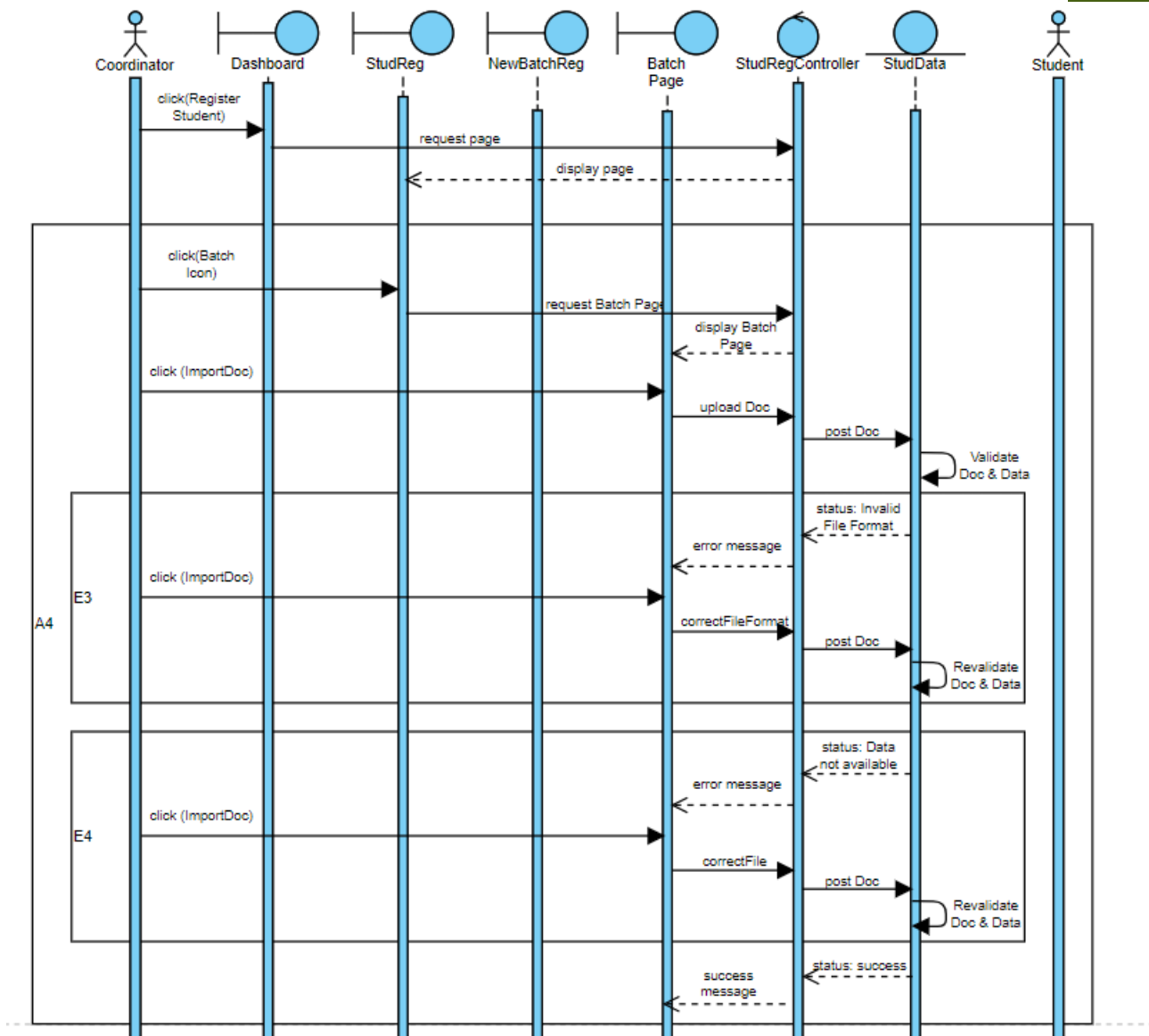
SOFTWARE REQUIREMENT SPECIFICATION (SRS)



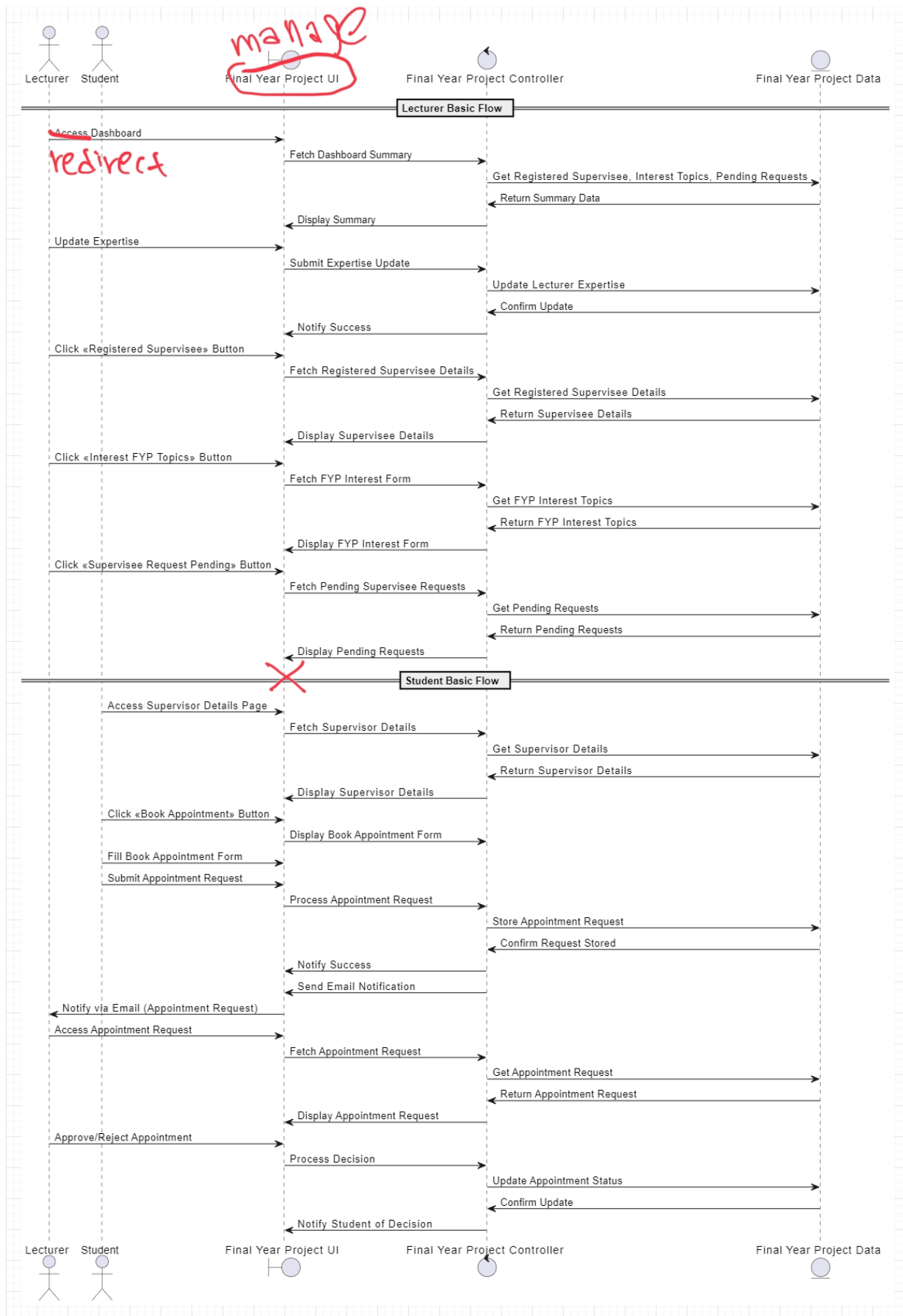
2.3 Exception flow (MSEF203_FYPSHA_2024)

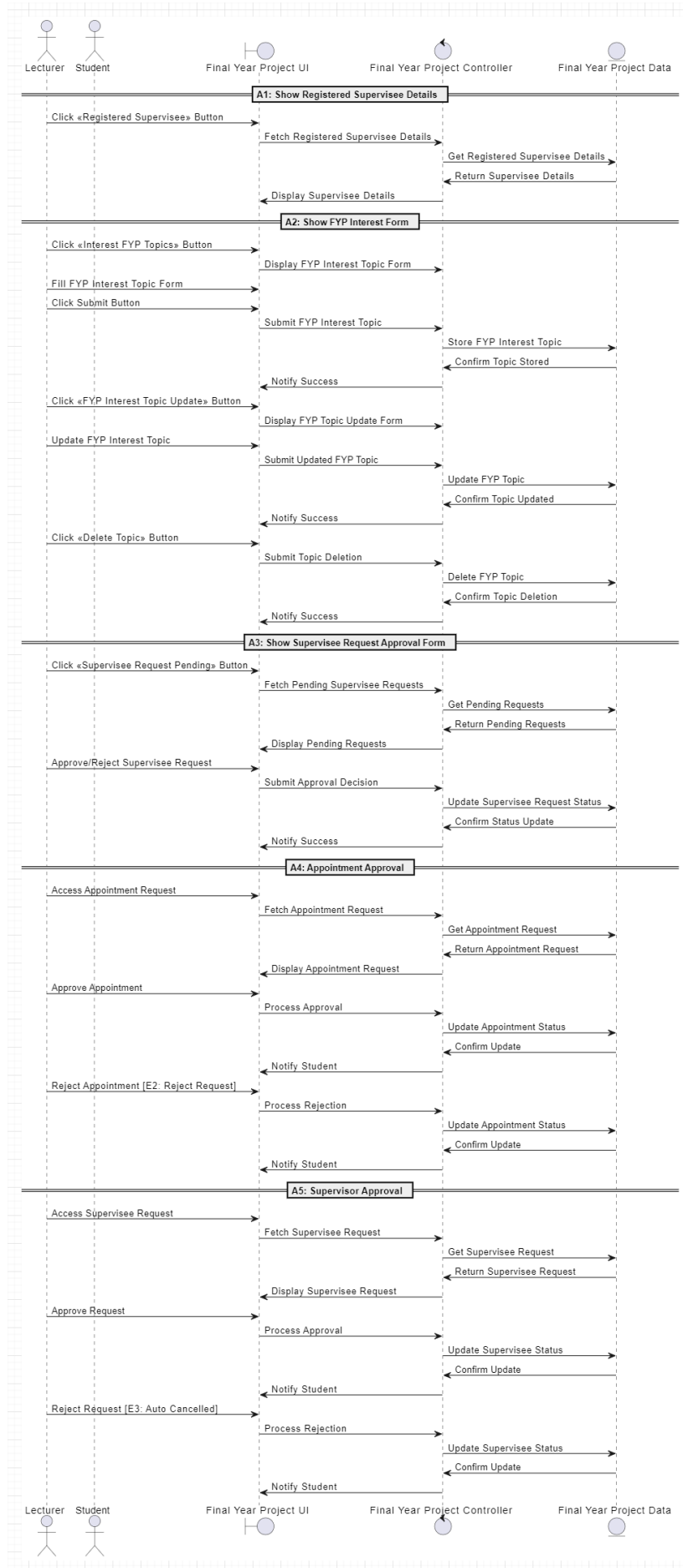


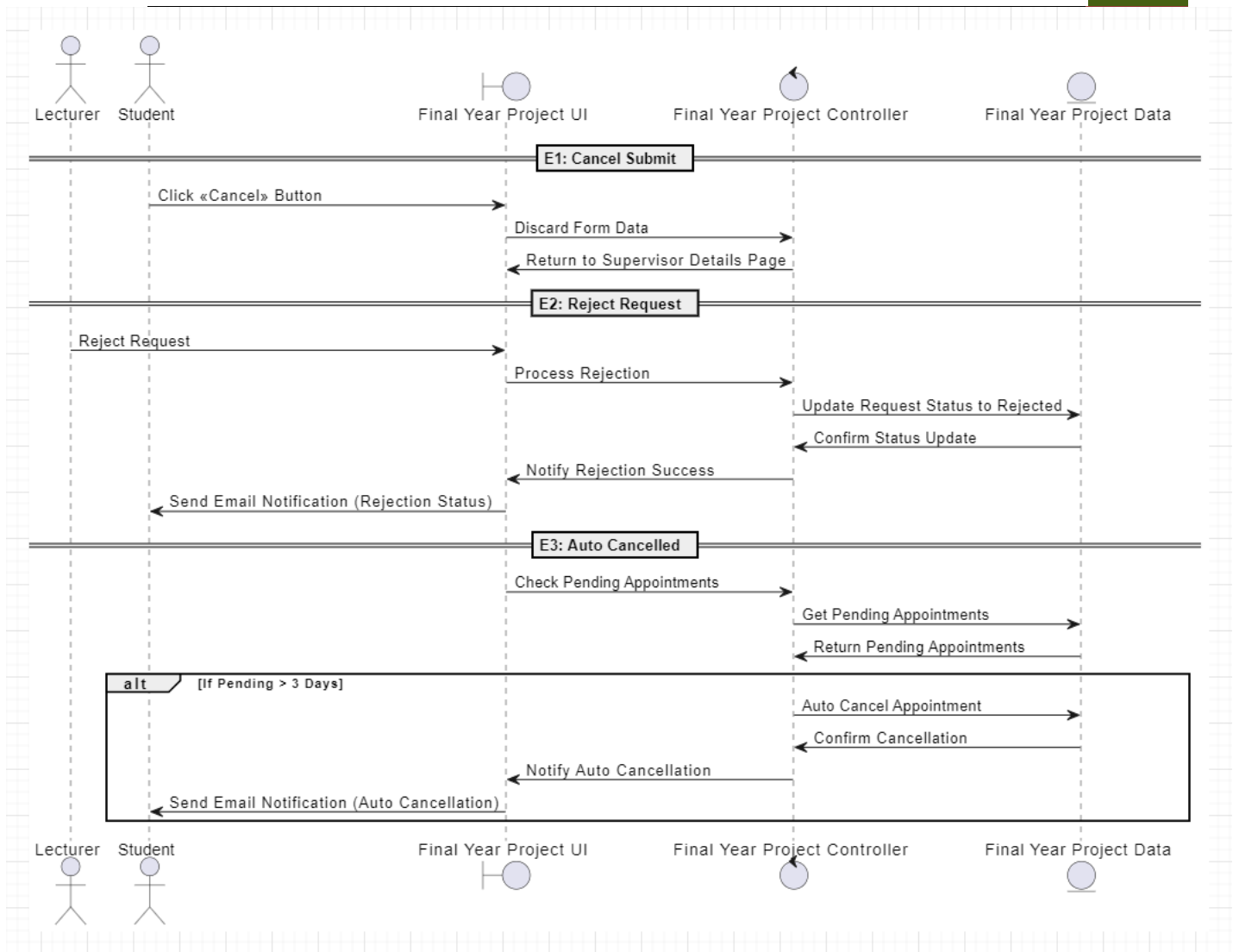
SOFTWARE REQUIREMENT SPECIFICATION (SRS)



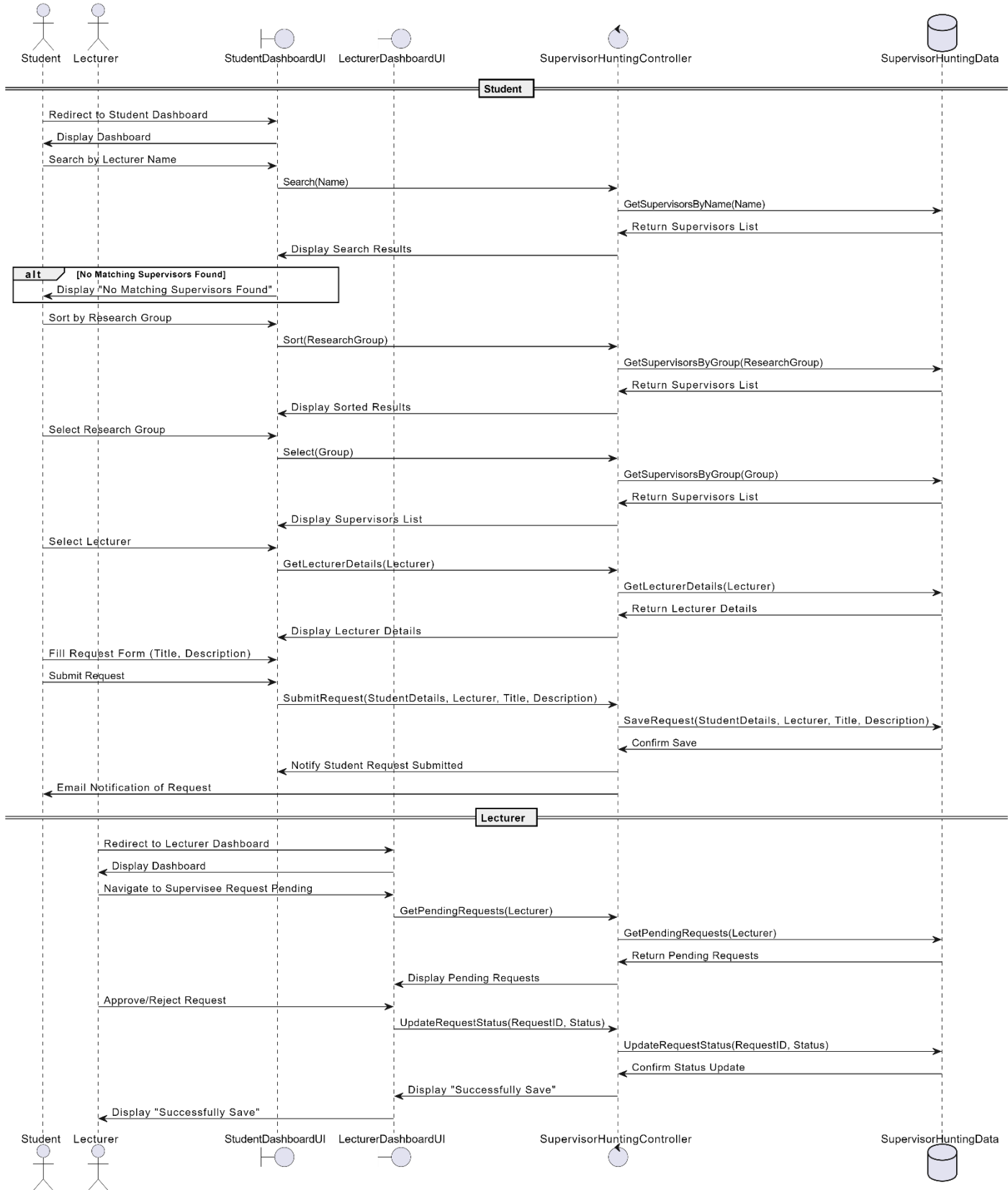
2. Manage Final Year Project (Mirza Rusyaidi)



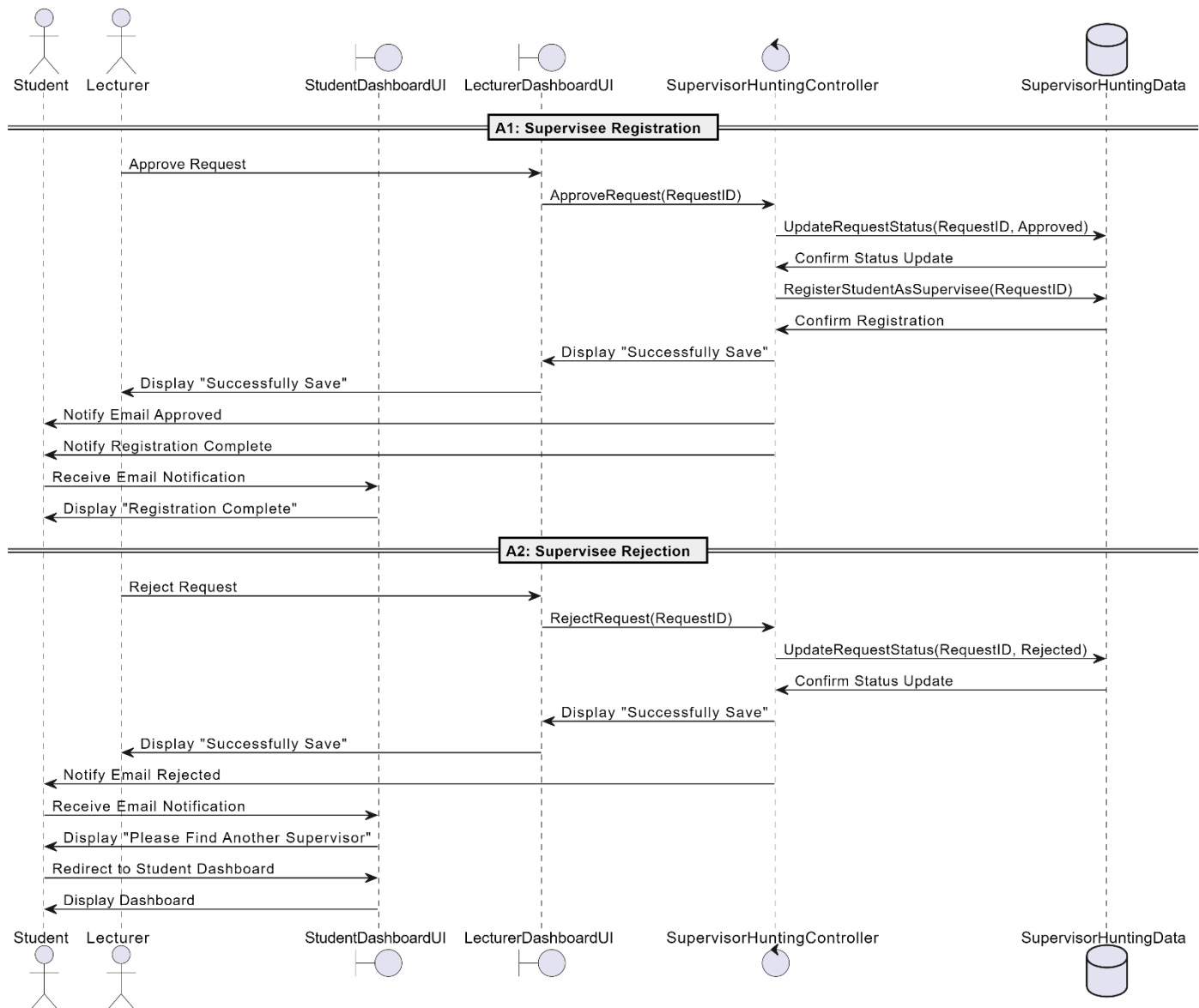




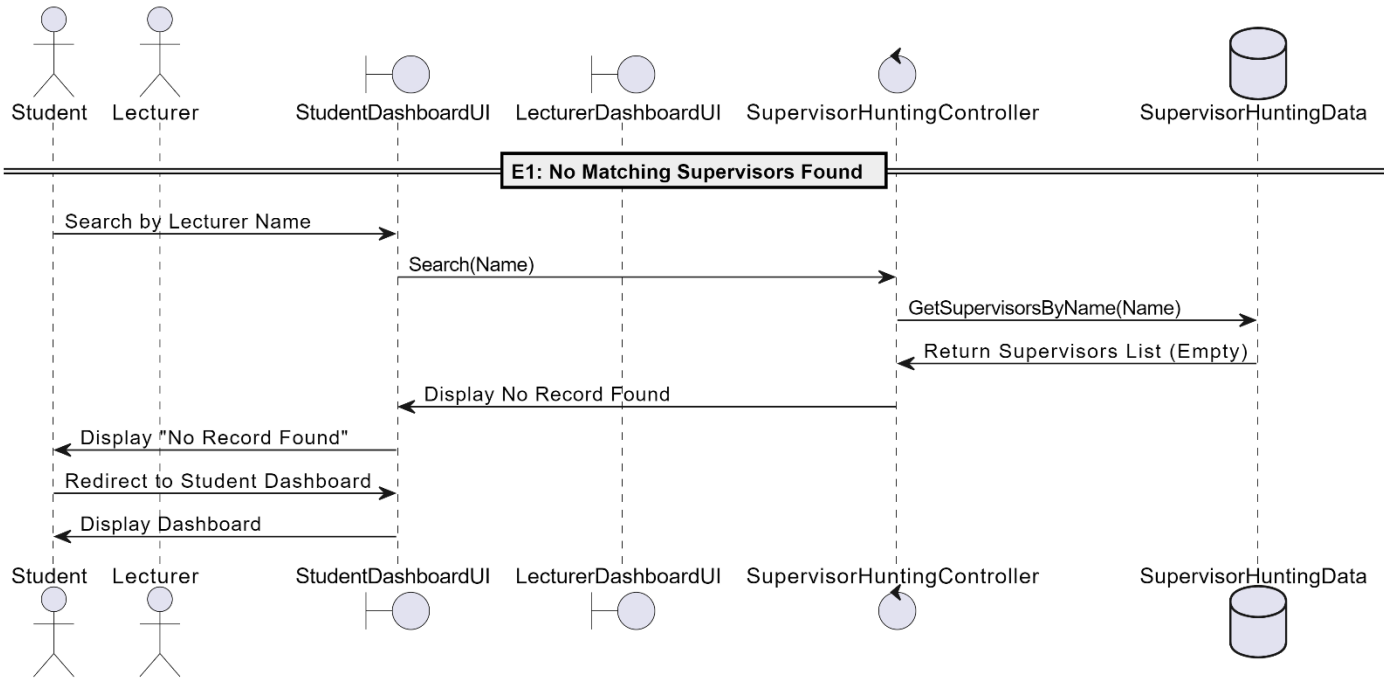
4.1 Basic Flow (MSHBF401_FYP_SHA_2024)



4.2 Alternative Flow (MSHBF402_FYP_SHA_2024)



4.3 Alternative Flow (MSHBF403_FYPSHA_2024)



APPENDIX B

User Interfaces

1. Manage Coordinator Activities

- Coordinator activities

FYP Supervisor hunting application

HOMEABOUTSERVICESCONTACT

ANNOUNCEMENT

REMINDER

Attention to All Student, Please be noted that duration of choosing your FYP topics will end in One week

Coordinator name

ADD ANNOUNCEMENT

COORDINATOR NAME:

DURATION:

SUBJECT:

MESSAGE:

ADD FILE

PUBLISH

SET SCHEDULE

TEACHING & RESEARCH & SUPERVISOR PRE

DEVELOPMENT 1/4/2026 25/6/2026

REVIEW & MEETING 25/6/2026 31/6/2026

GENERATE REPORT

Supervisor Name	Number of Students Supervised	Activity Status
Dr. John Smith	5	Completed
Dr. Jane Doe	3	Pending
Dr. Alan Brown	4	Completed
Dr. Mary Johnson	2	Canceled
Dr. Emily Davis	6	Completed

- Add announcement

COORDINATOR NAME:

DURATION:

SUBJECT:

MESSAGE:

ADD FILE

PUBLISH

Schedule

SUCCESSFULLY SAVE REPORT
FILE



Report on Supervisor hunting system

Date of Activity	Supervisor Name	Number of Students Supervised	Activity Status	Comments
2024-06-01	Dr. John Smith	5	Completed	None
2024-06-02	Dr. Jane Doe	3	Pending	Awaiting feedback
2024-06-03	Dr. Alan Brown	4	Completed	Successful session
2024-06-04	Dr. Mary Johnson	2	Canceled	Rescheduled
2024-06-05	Dr. Emily Davis	6	Completed	None

GENERATE REPORT

SAVE

FYP Supervisor hunting application

HOMEABOUTSERVICESCONTACT

FYP TOPIC SEARCHING REVIEW
25/3/2024

RESEARCH OF SUPERVISOR
1/4/2024 - 5/4/2024

SUPERVISOR HUNTING BEGIN
6/4/2024 - 20/4/2024

SUPERVISOR HUNTING END
1/5/2024

DELETE SCHEDULE

FYP TOPIC SEARCHING & MEETING:

RESEARCH & DEVELOPMENT:

SUPERVISOR REVIEW & MEETING:

PRESENTATION:

ADD SCHEDULE

• Generate Reports

FYP Supervisor hunting application

HOMEABOUTSERVICESCONTACT

SUCCESSFULLY SAVE REPORT FILE

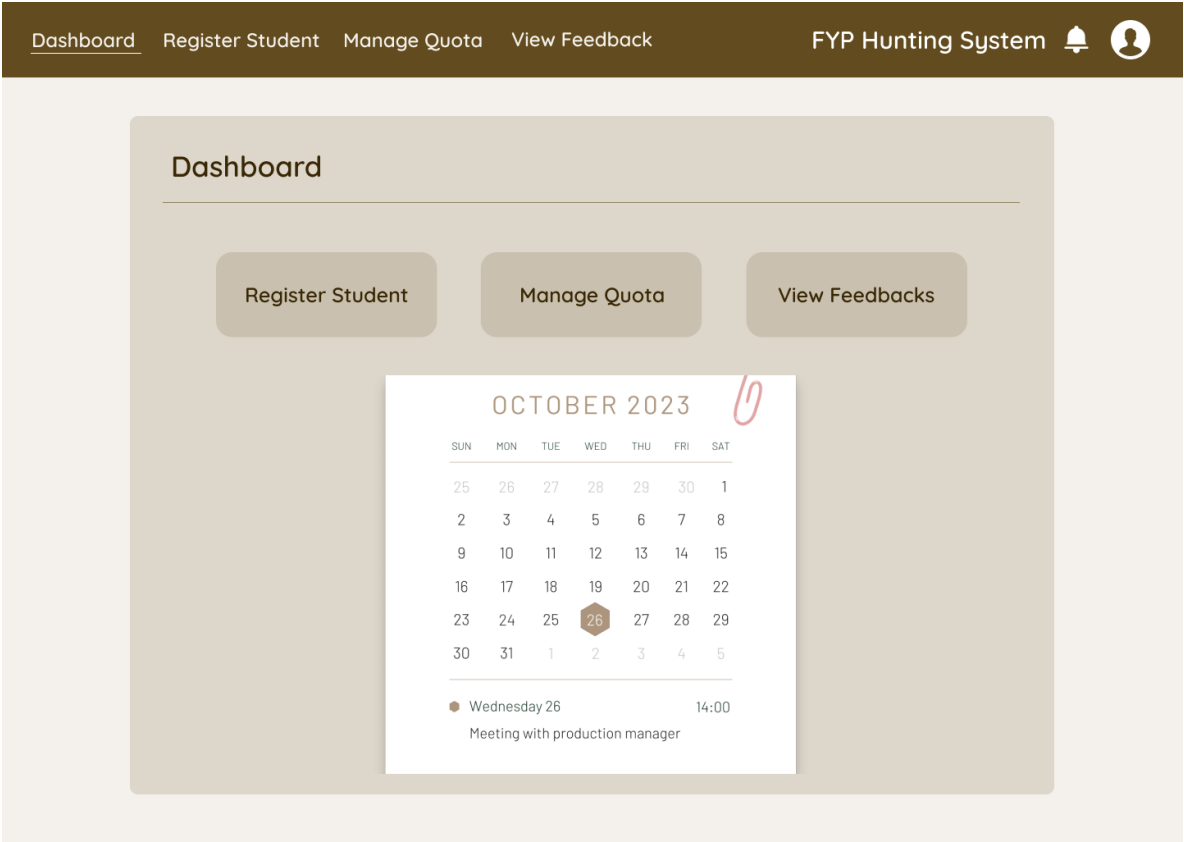
Report on Supervisor hunting system

Date of Activity	Supervisor Name	Number of Students Supervised	Activity Status	Comments
2024-06-01	Dr. John Smith	5	Completed	None
2024-06-02	Dr. Jane Doe	3	Pending	Awaiting feedback
2024-06-03	Dr. Alan Brown	4	Completed	Successful session
2024-06-04	Dr. Mary Johnson	2	Canceled	Rescheduled
2024-06-05	Dr. Emily Davis	6	Completed	None

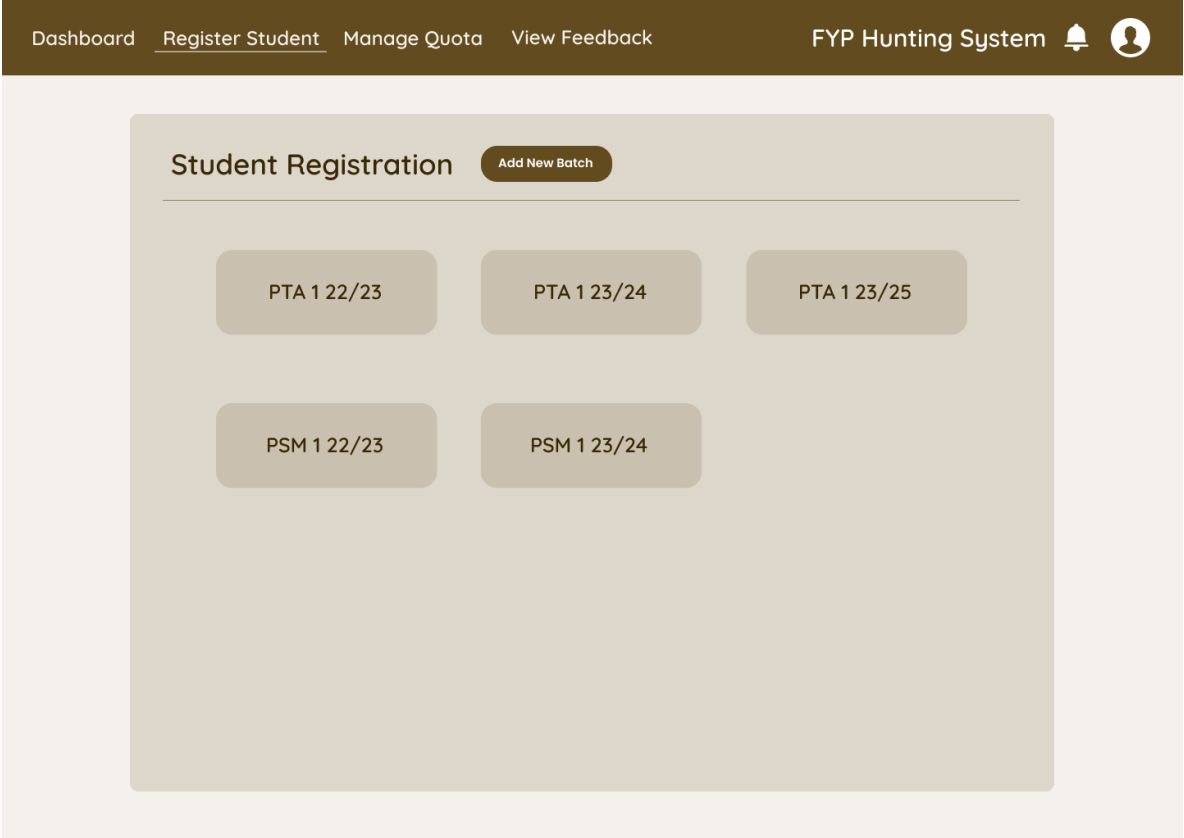
GENERATE REPORT

SAVE

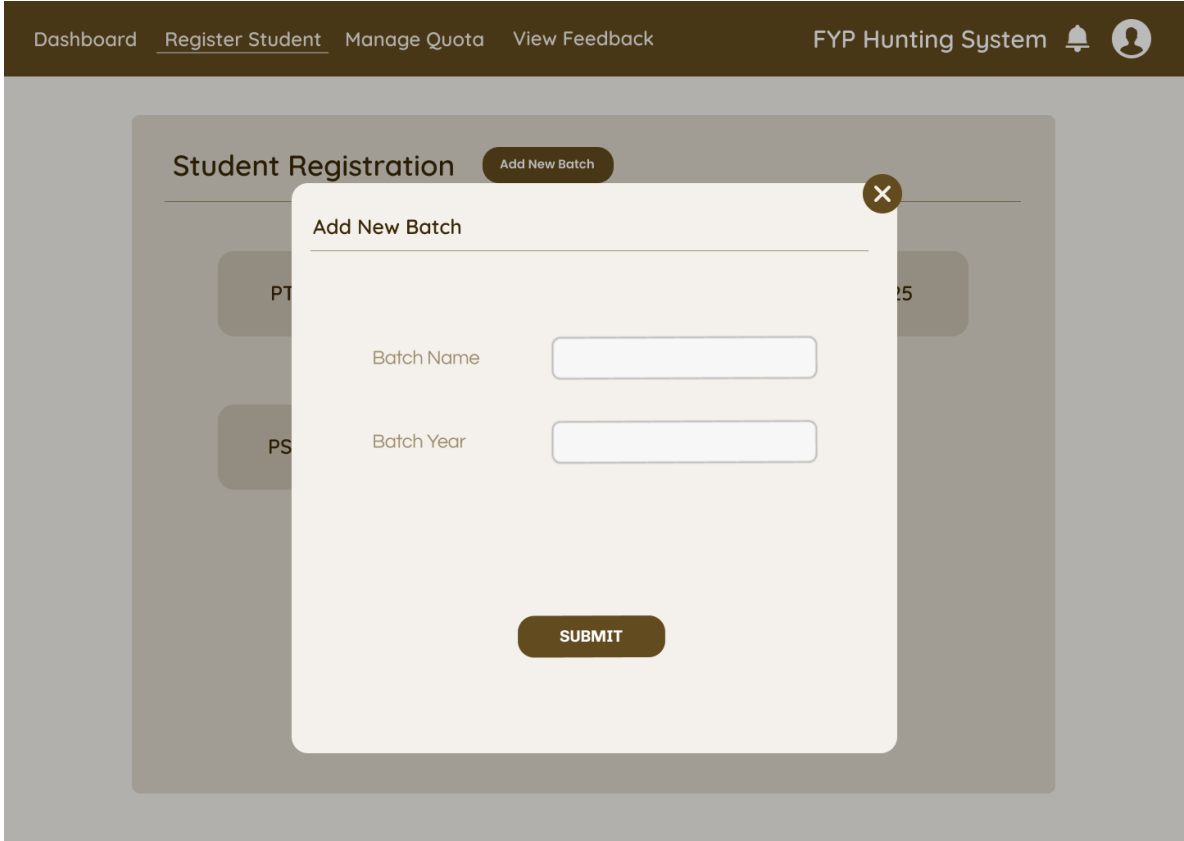
2. Manage Student (Amala Karthigayan)
Coordinator Dashboard:



Student Registration Page



Add New Batch Page



Student List Page

DashboardRegister StudentManage QuotaView Feedback

FYP Hunting System

PTA 1 22/33

Import DocumentGenerate Credential

Name	Matric ID	User ID	Password
Jane	CB23116	0302129034	*****
Jane	CB23116	0302129034	*****
Jane	CB23116	0302129034	*****
Jane	CB23116	0302129034	*****
Jane	CB23116	0302129034	*****
Jane	CB23116	0302129034	*****

Supervisee Quota

DashboardRegister StudentManage QuotaView Feedback

FYP Hunting System

Manage Quota

Name	Supervisor ID	Contact	Status	Quota	
Jane	CB23116	0302129034	Active	-	+
Jane	CB23116	0302129034	Active	2	
Jane	CB23116	0302129034	Active	5	
Jane	CB23116	0302129034	Active	-	+
Jane	CB23116	0302129034	Active	1	
Jane	CB23116	0302129034	Active	2	

View Feedback

DashboardRegister StudentManage QuotaView Feedback

FYP Hunting System



View Feedback

Name	ID	Contact	Feedback
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 
Jane	CB23116	0302129034	Lorem Ipsum is simply dummy text of the printing and typesetting industry. 

Print

3. Manage Final Year Project (Mirza Rusyaidi & Nur Nabilah)
Lecturer Dashboard

LOGO

FYP SUPERVISOR HUNTING APPLICATION



Registered Supervisee

FYP Interest Form

Supervisee Request Pending

Supervisee Details

LOGO

FYP SUPERVISOR HUNTING APPLICATION

STUDENT DETAILS

	Student Name	Matric ID	FYP Title

Final Year Project Interest Topic

LOGO

FYP SUPERVISOR HUNTING APPLICATION

STUDENT DETAILS

	Student Name	Matric ID	FYP Title

LOGO

FYP SUPERVISOR HUNTING APPLICATION**FYP Interest Topic
Form**

Title

Description

SUBMIT

FYP Interest Form Update

LOGO

FYP SUPERVISOR HUNTING APPLICATION**FYP Interest Topic
Form**

Title

Description

UPDATE

Supervisee Request Approval Form (Pending)

LOGO

FYP SUPERVISOR HUNTING APPLICATION

STUDENT DETAILS

	Title	Brief Description	Status	
	Student Name		Pending	

Supervisee Request Approval Form (Approved)

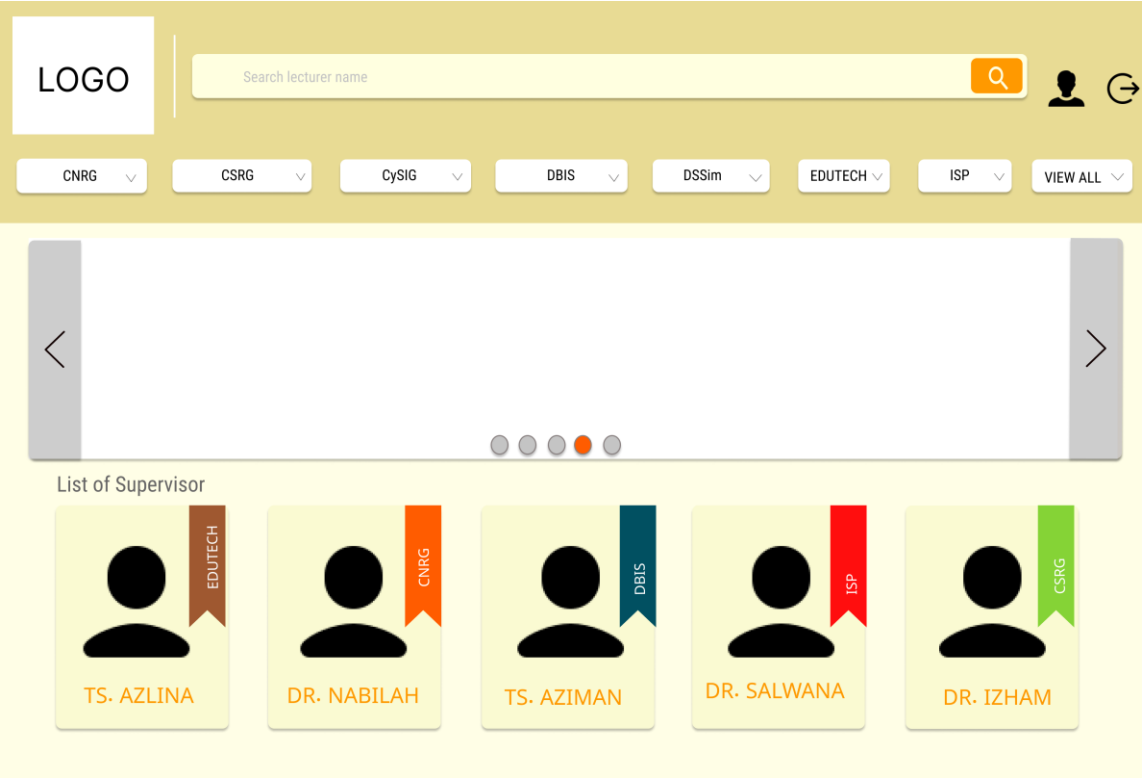
LOGO

FYP SUPERVISOR HUNTING APPLICATION

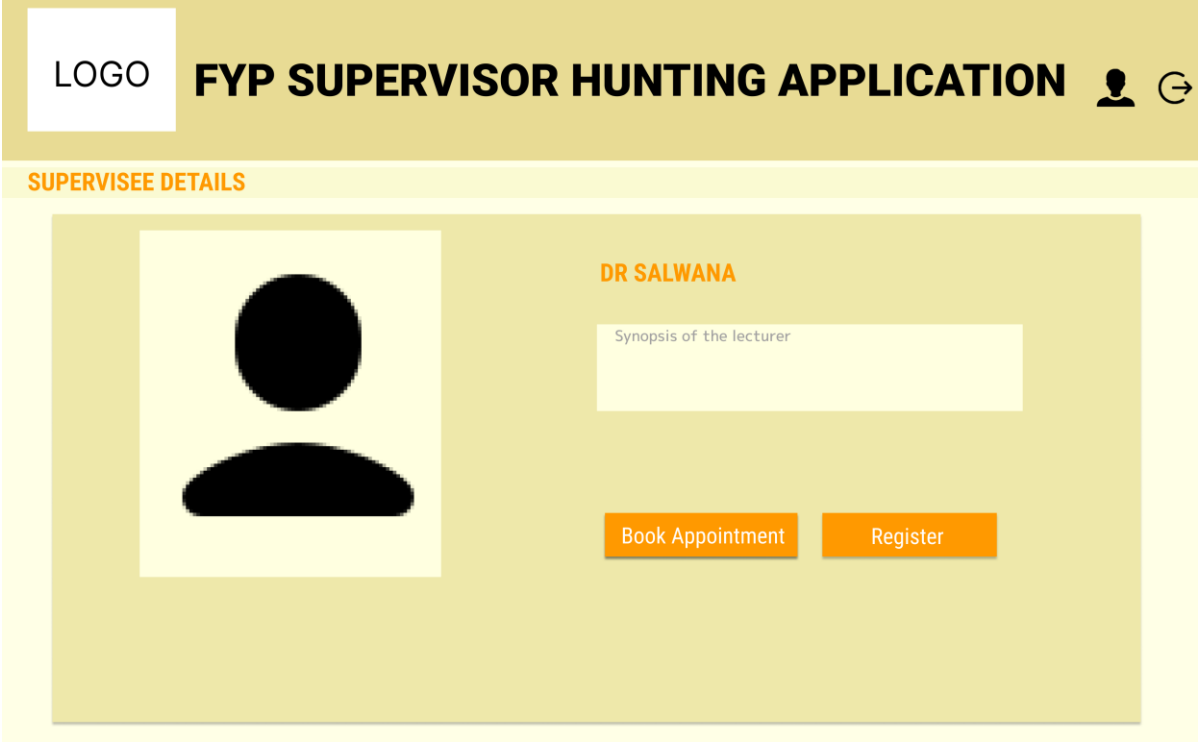
STUDENT DETAILS

	Title	Brief Description	Status	
	Student Name		Approved	

Student Dashboard



Supervisor Details



Book Appointment

LOGO

FYP SUPERVISOR HUNTING APPLICATION

Appointment

Date

Time

Reason

SUBMIT

Supervisee Registration

LOGO

FYP SUPERVISOR HUNTING APPLICATION

Register

Title

Brief Description

SUBMIT

Sort Research Group

LOGO

Search lecturer name

Q



CNRG

CSRG

CySIG

DBIS

DSSim

EDUTECH

ISP

VIEW ALL

<

>

List of Supervisor



EDUTECH

TS. AZLINA



CNRG

DR. NABILAH



DBIS

TS. AZIMAN



ISP

DR. SALWANA



CSRG

DR. IZHAM

Search Lecturer Name

LOGO

AZLINA

Q



CNRG

CSRG

CySIG

DBIS

DSSim

EDUTECH

ISP

VIEW ALL



EDUTECH

TS. AZLINA

SRS FYPSHA 2024 VERSION 1.1.24

Student Profile

LOGO

FYP SUPERVISOR HUNTING APPLICATION

Student Profile

Name

Matric ID

Email

Password

Phone Number

UPDATE

Lecturer Profile

LOGO

FYP SUPERVISOR HUNTING APPLICATION

Lecturer Profile

Name

Staff ID

Email

Password

Phone Number

Expertise

UPDATE